Túngara frog

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Tungara Frog

Engystomops pustulosus



1

fringe-lipped bat



habitat



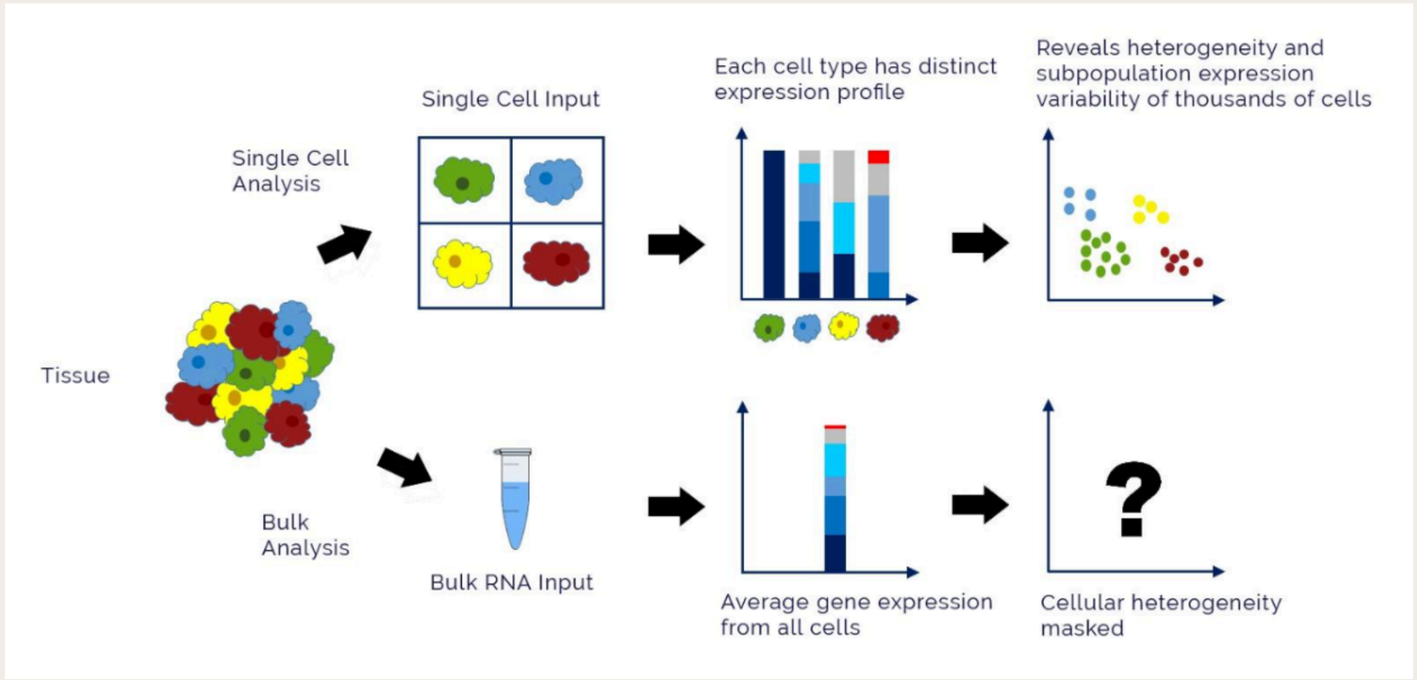




- **Courtship Calls**
  - Whine / Whine → Chuck (multiple)
- **Size**
  - ~3.3–3.6 cm
- **Range**
  - Southern Mexico → northern South America
- **Habitat**
  - ponds near forest
- **Activity**
  - Nocturnal, calls all year,
- **Nest**
  - Floating foam nests on the water surface

DATA OVERVIEW

The dataset uses single-**nuclei** RNA sequencing from female túngara frogs exposed to whine-chuck calls, whine calls, or green noise. By measuring gene expression at the individual-cell level, the data allow us to identify distinct brain cell populations and examine how they respond during mate-choice decisions.



Question

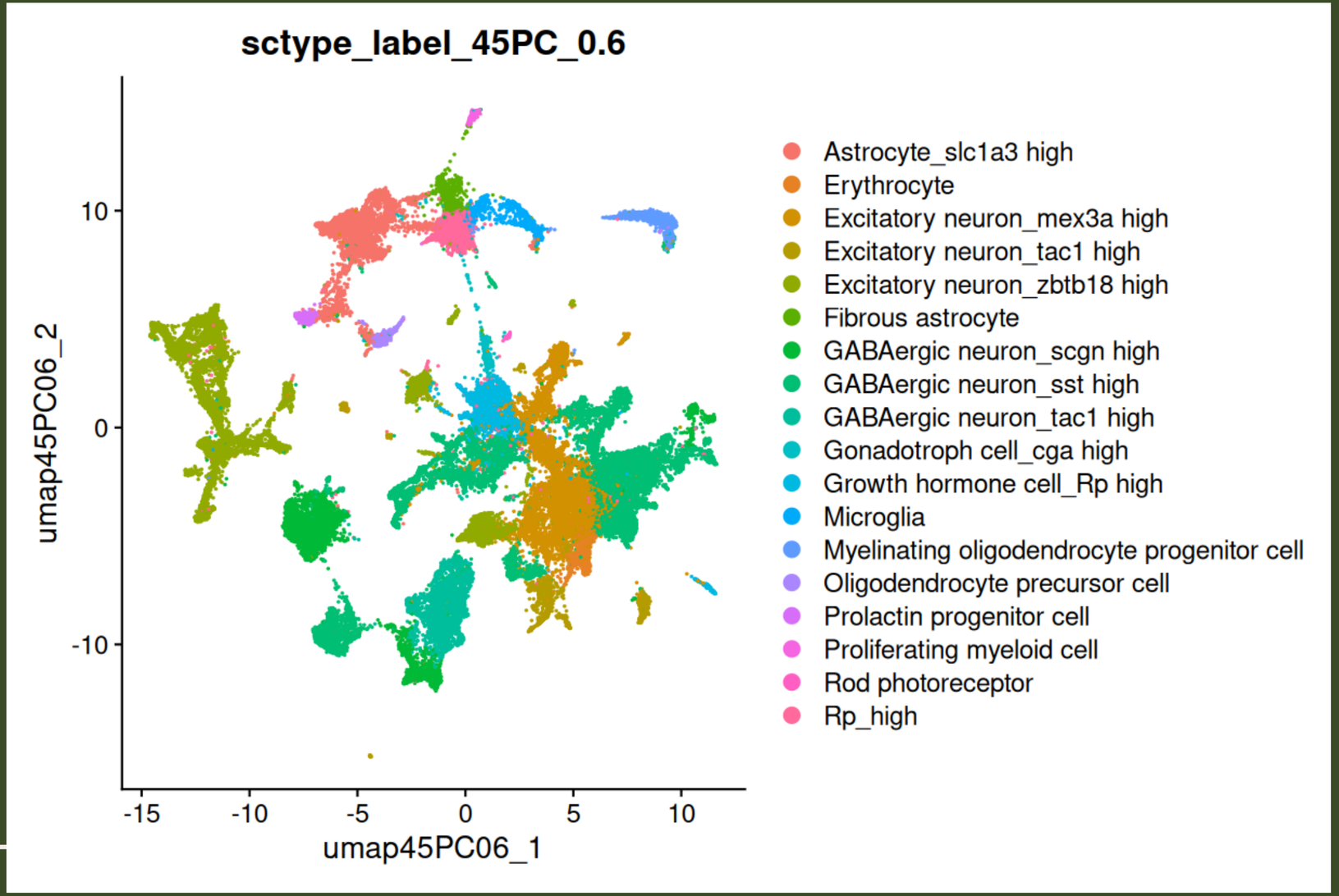
Which genes and pathways are upregulated in response to the Whine-Chuck noise?

3

Annotation

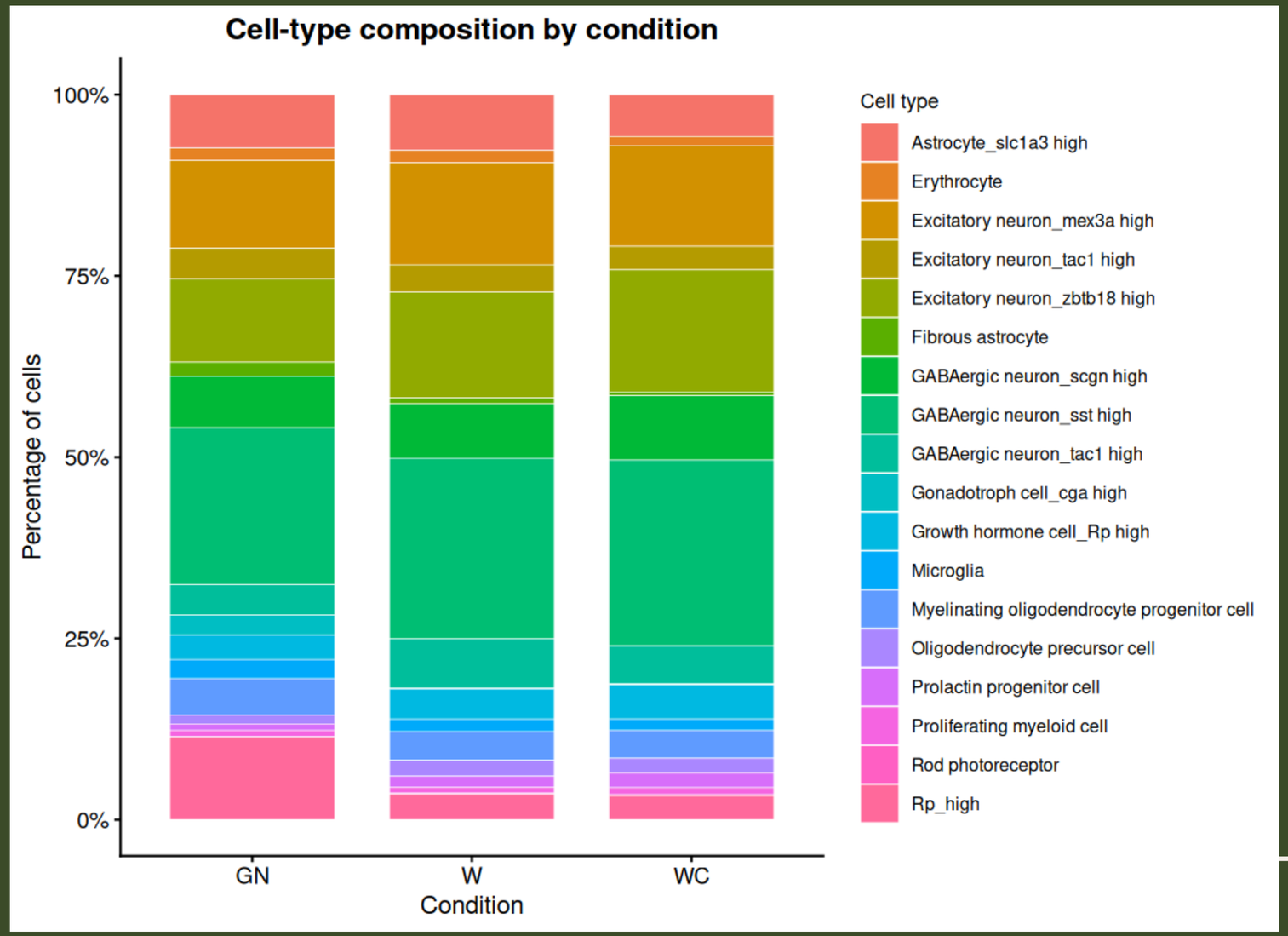
UMAP Orthofinder

To annotate the dataset by cell type, we created a custom brain marker set using marker genes from the Xenopus Cell Landscape. These markers were then adapted for the túngara frog dataset and used to assign cell-type identities to the clusters using SCtype.



Cell Type Inventories

BY CONDITION

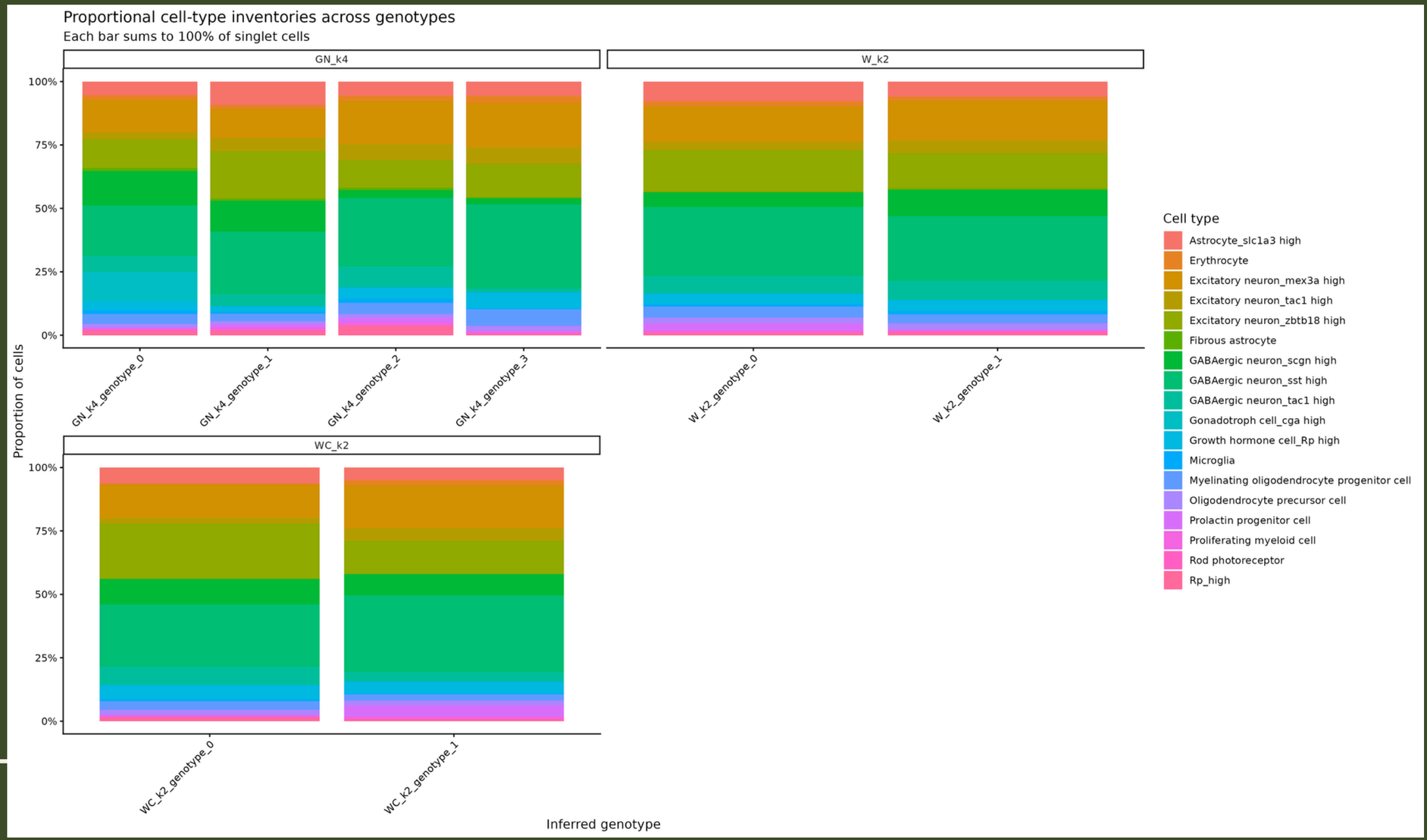


Cell Count

GN	W	WC
18809	17865	21798

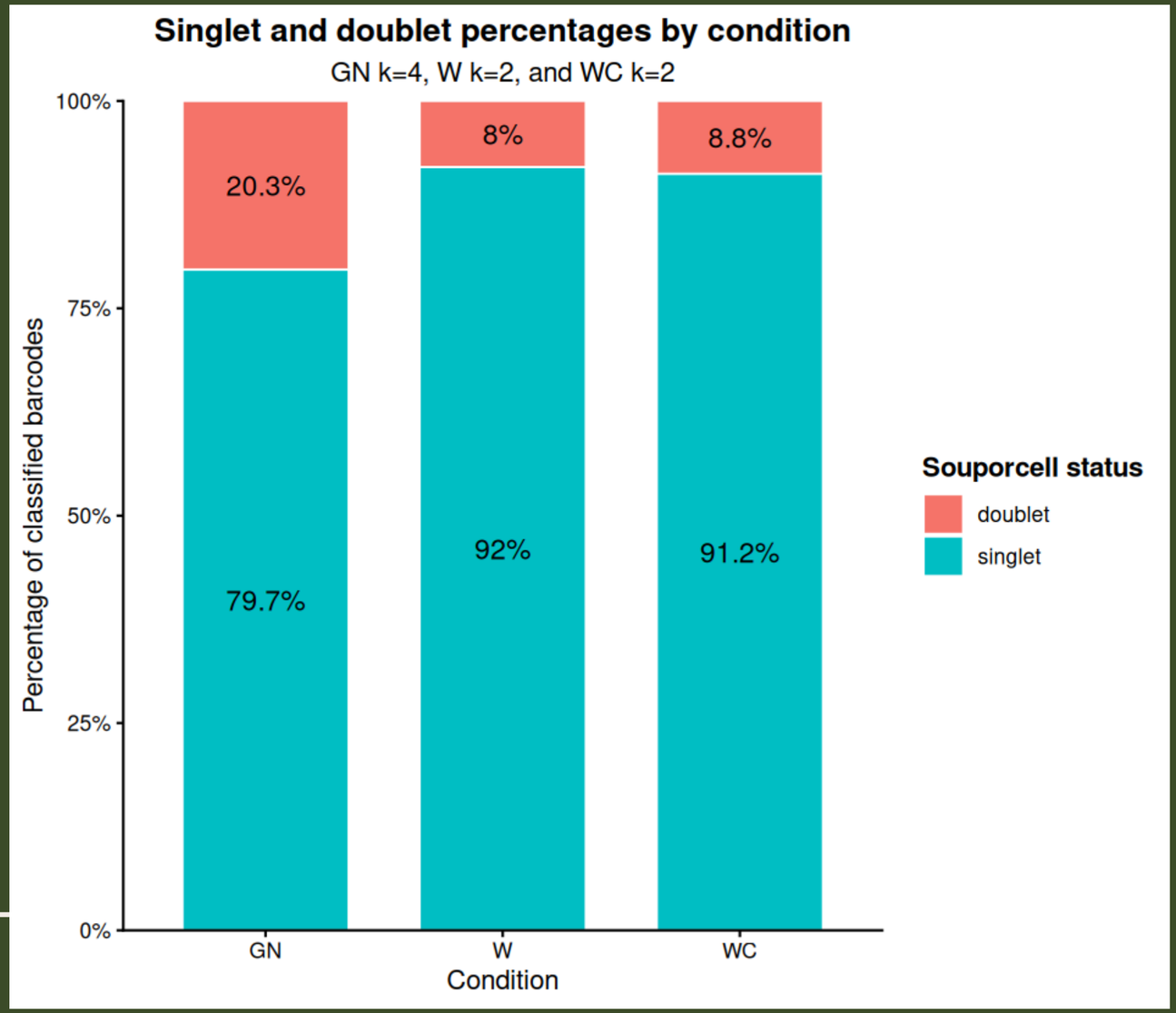
Cell Type Inventories

BY GENOTYPES

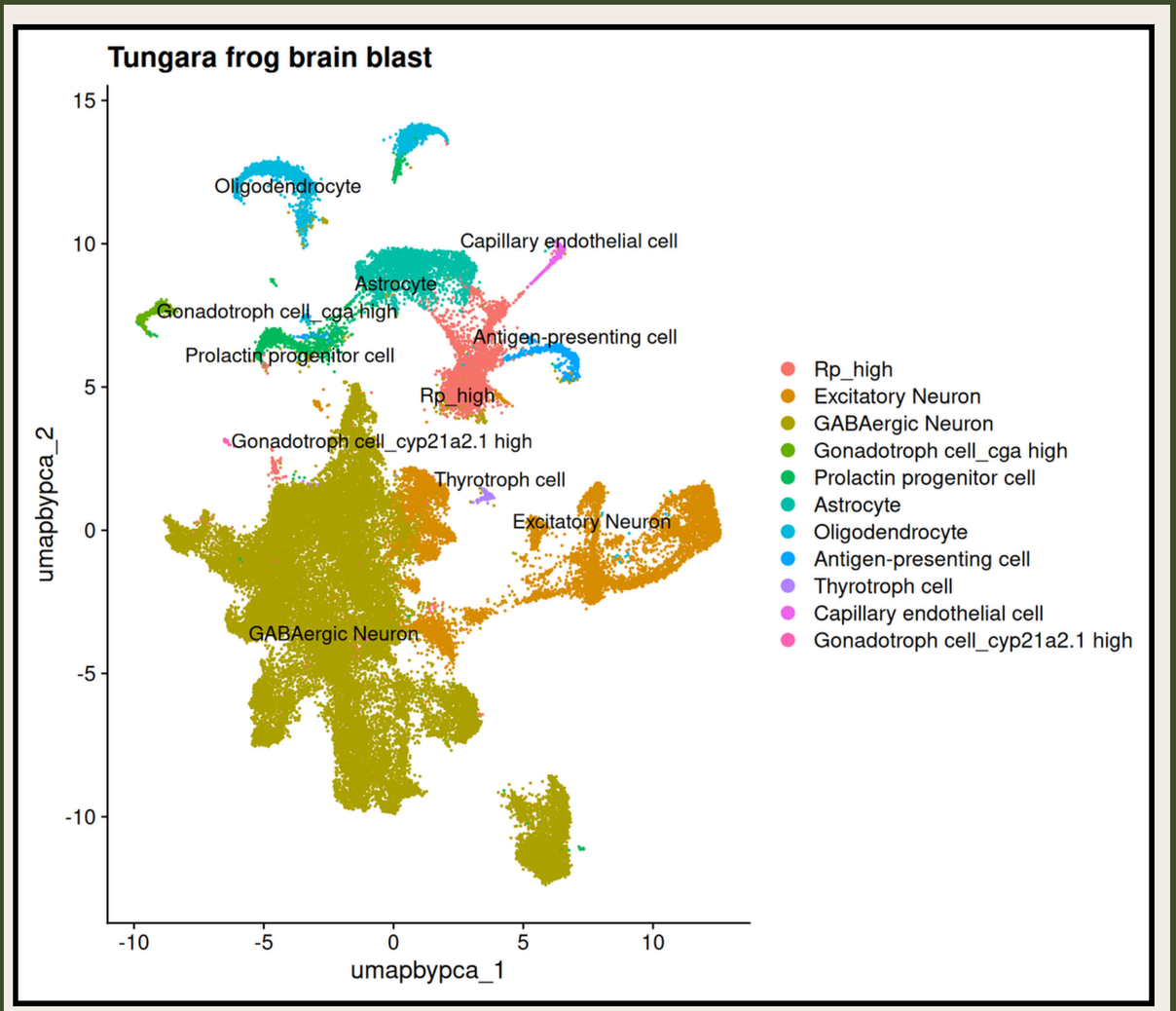


SINGLETS & DOUBLETS

BY CONDITIONS



Blast Annotation

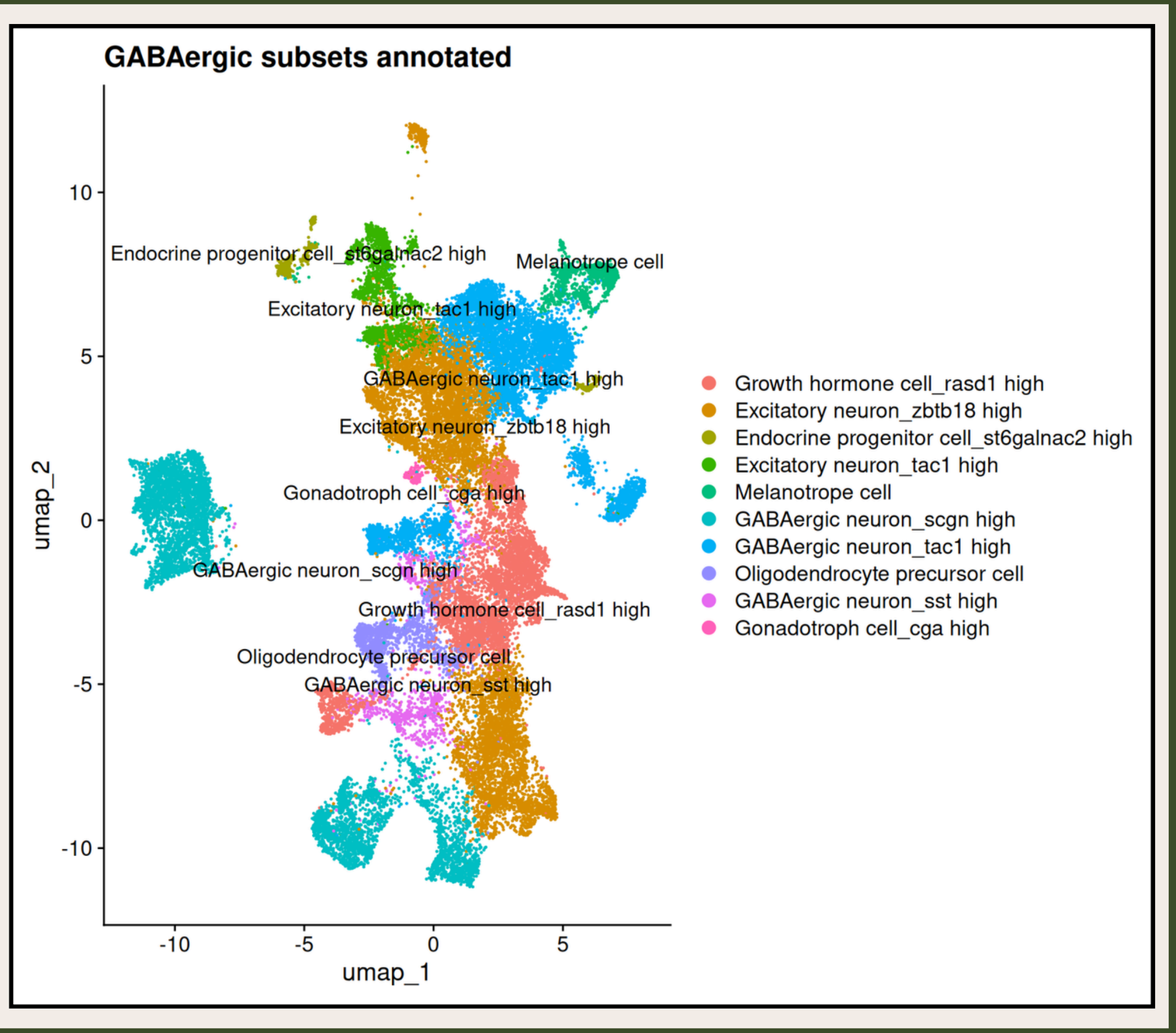


- **Cross-Species Ortholog Mapping**
  - Reciprocal Best Hit BLAST (Xenopus ↔ Túngara)
- **sc-type Automated Annotation**
  - Xenopus brain markers to score and identify Seurat clusters
- **Subclustering And Refinement**
  - GABAergic neurons re-clustered at high resolution and mapped subtypes back to the dataset.

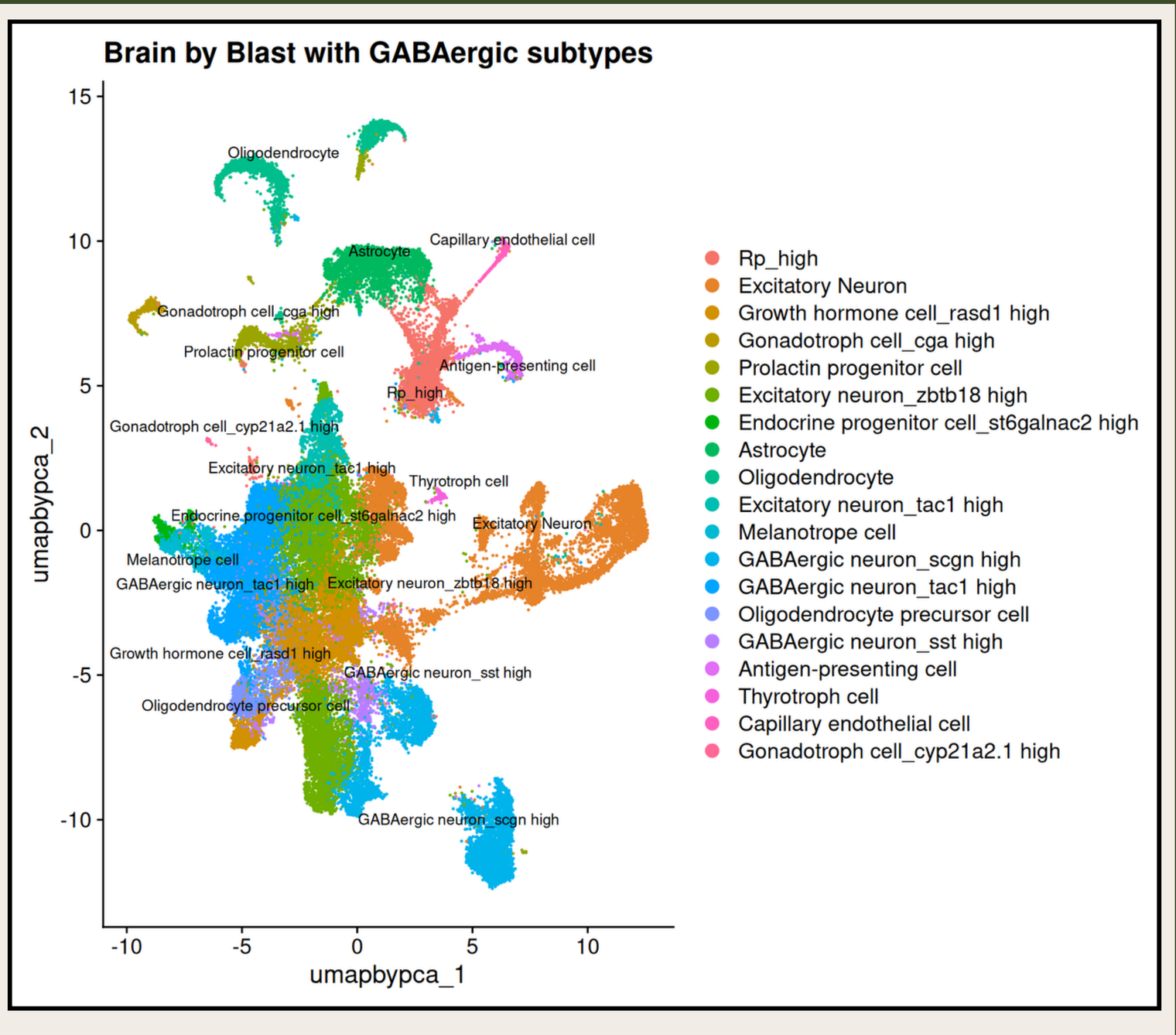
Key annotation numbers

- **19 541 raw protein-to-protein matches between the species**
- **15 981 translatable gene names**
- **Over 26 000 mapped to known cell-type marker**
- **854 essential marker genes isolated**

Gabaergic broken down



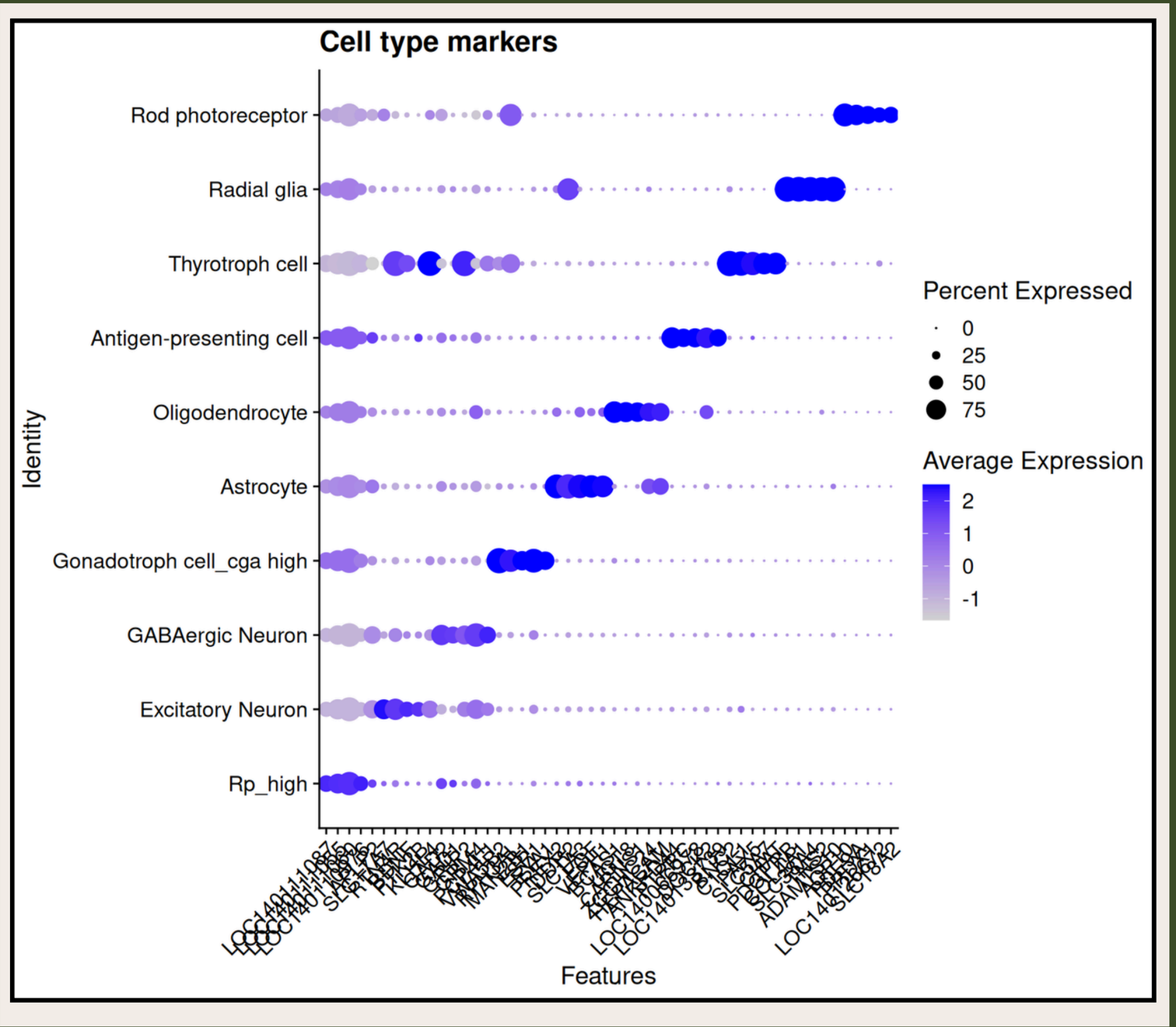
Final annotation



Cell breakdown in %

Antigen-presenting cell	Astrocyte_slc1a3 high	Capillary endothelial cell
1.46	5.28	0.90
Endocrine progenitor cell_st6galnac2 high	Excitatory neuron_mex3a high	Excitatory neuron_tac1 high
0.88	0.28	7.56
Excitatory neuron_zbtb18 high	GABAergic neuron_scgn high	GABAergic neuron_sst high
27.40	10.48	1.77
GABAergic neuron_tac1 high	Gonadotroph cell_cga high	Gonadotroph cell_cyp21a2.1 high
12.23	1.25	0.10
Growth hormone cell_rasd1 high	Melanotrope cell	Myelinating oligodendrocyte progenitor cell
9.58	1.78	4.52
Oligodendrocyte precursor cell	Prolactin progenitor cell	Rp_high
4.40	2.91	6.88
Thyrotroph cell		
0.34		

Cell markers coherence across cell type

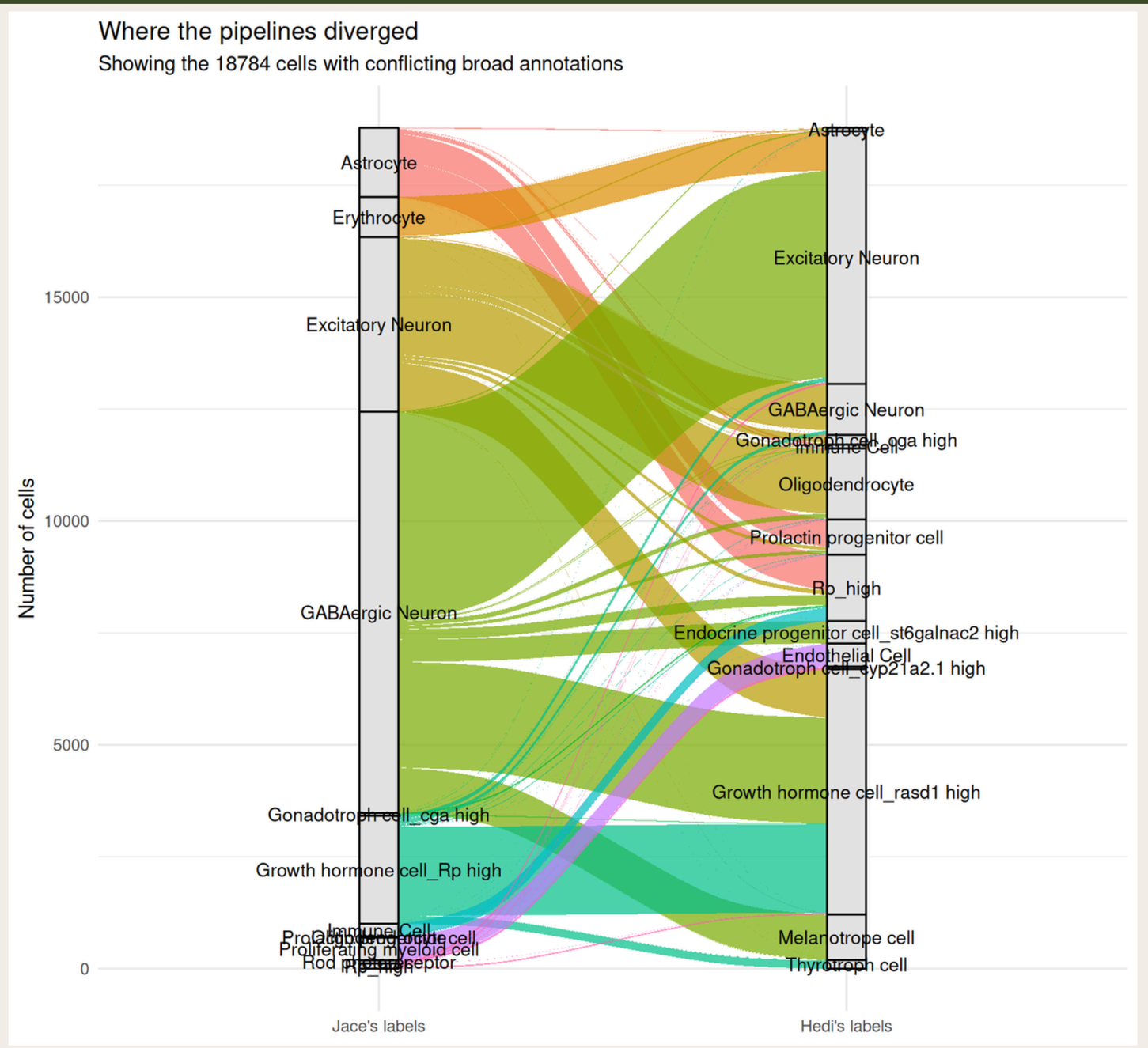


Method Comparaison

Agreement: 70.97%

Adjusted rand index: 0.344

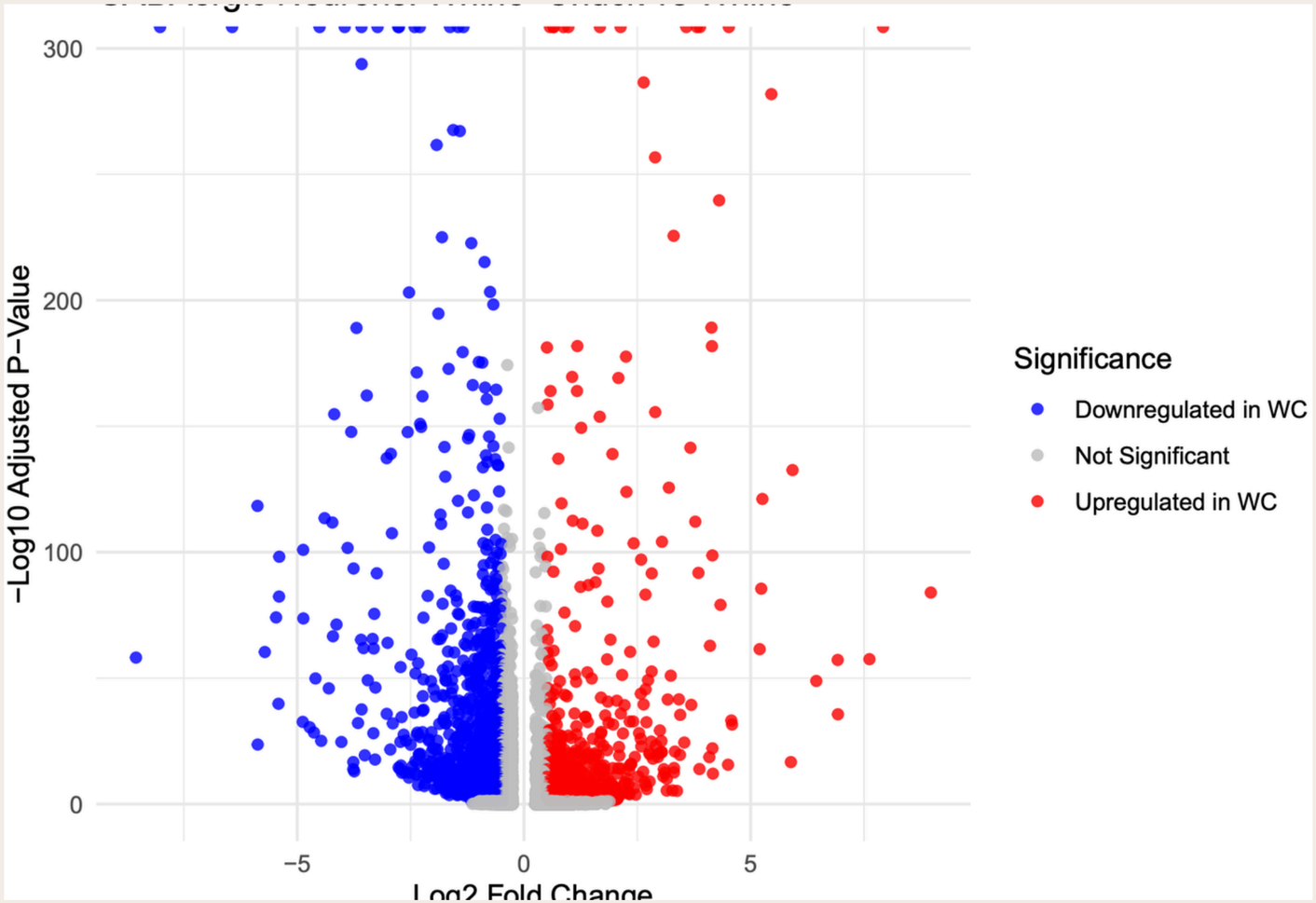
Global Agreement: 94.69%



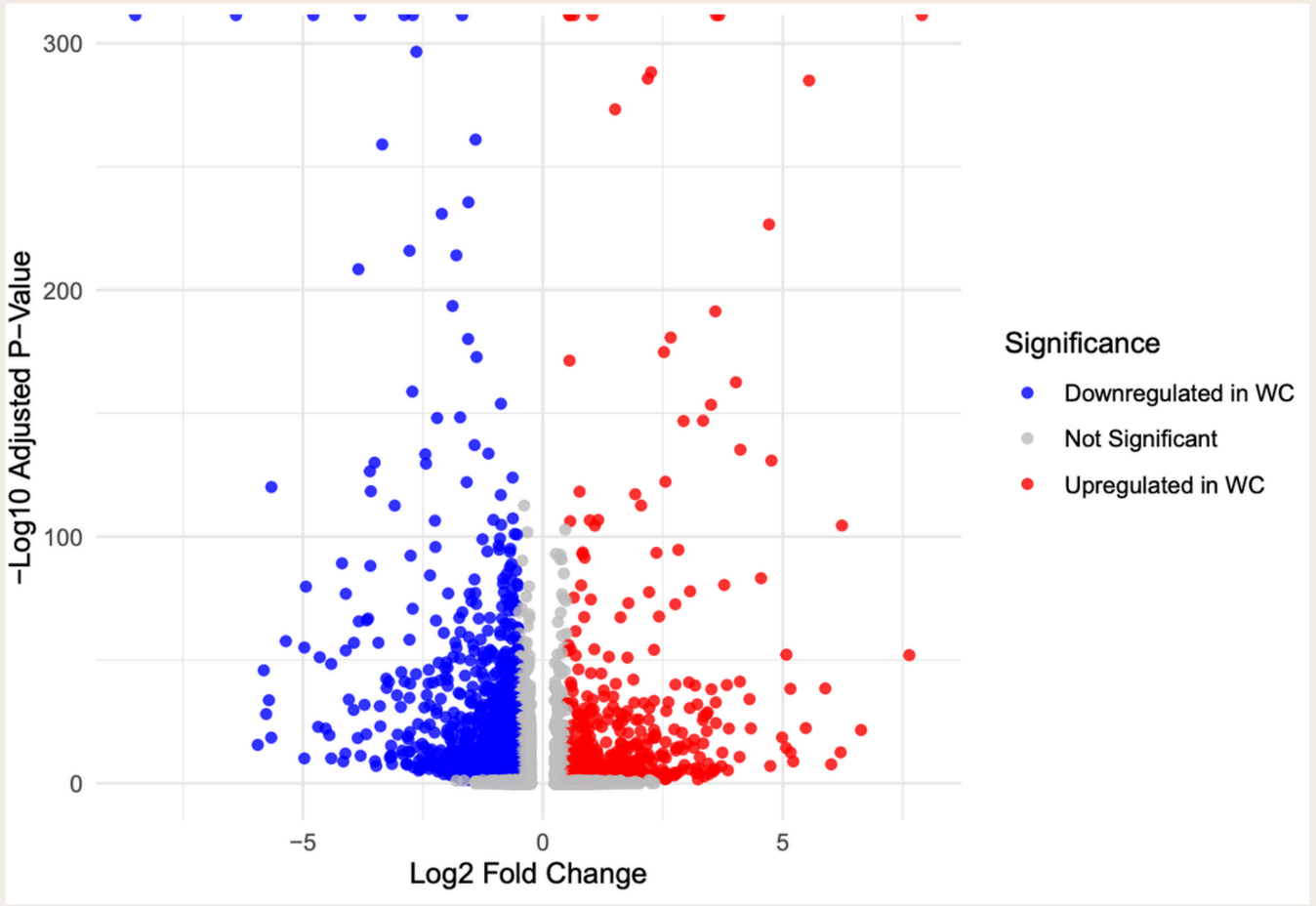
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DEG ANALYSIS

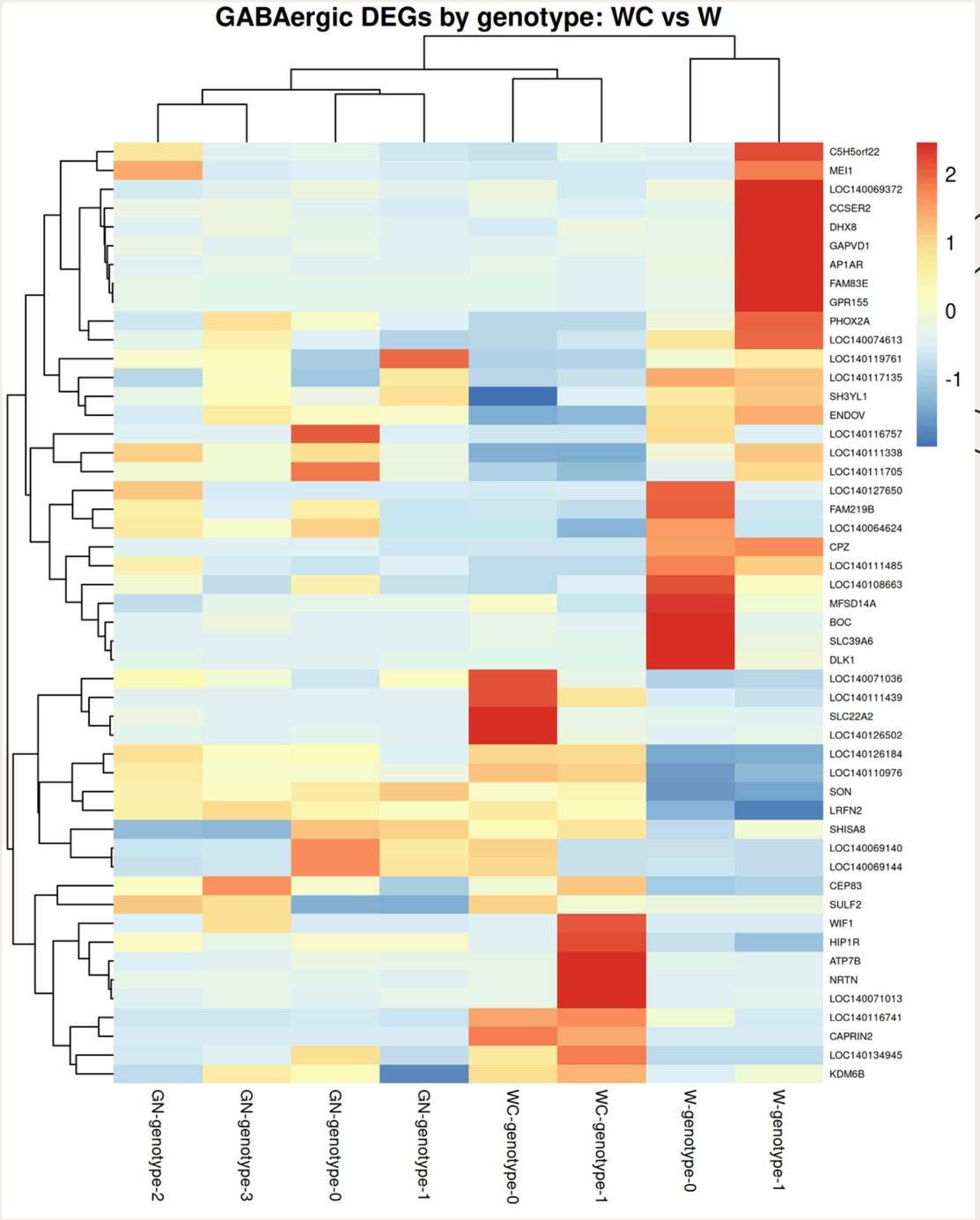
GABAERGIC - WC vs. W



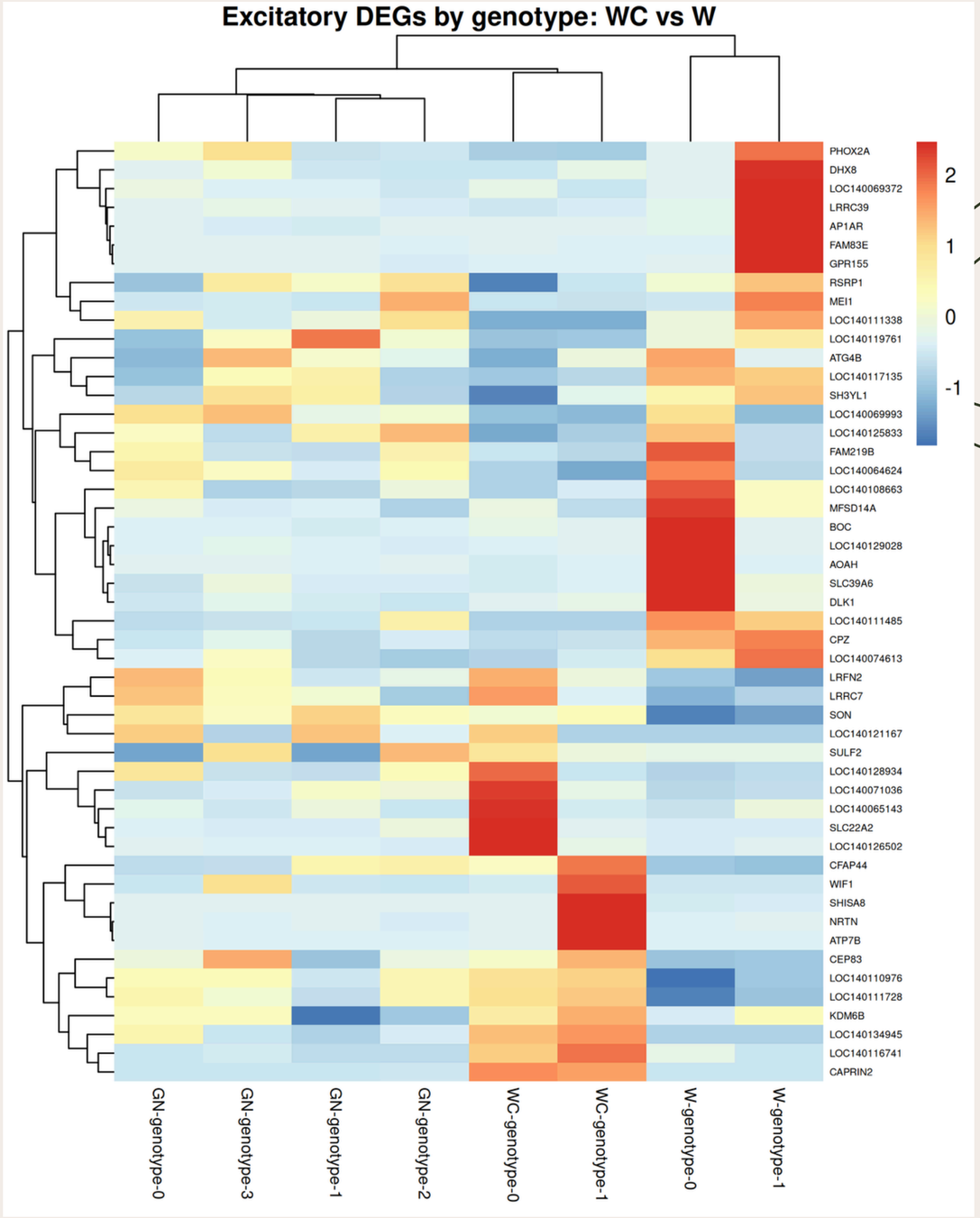
EXCITATORY - WC vs. W



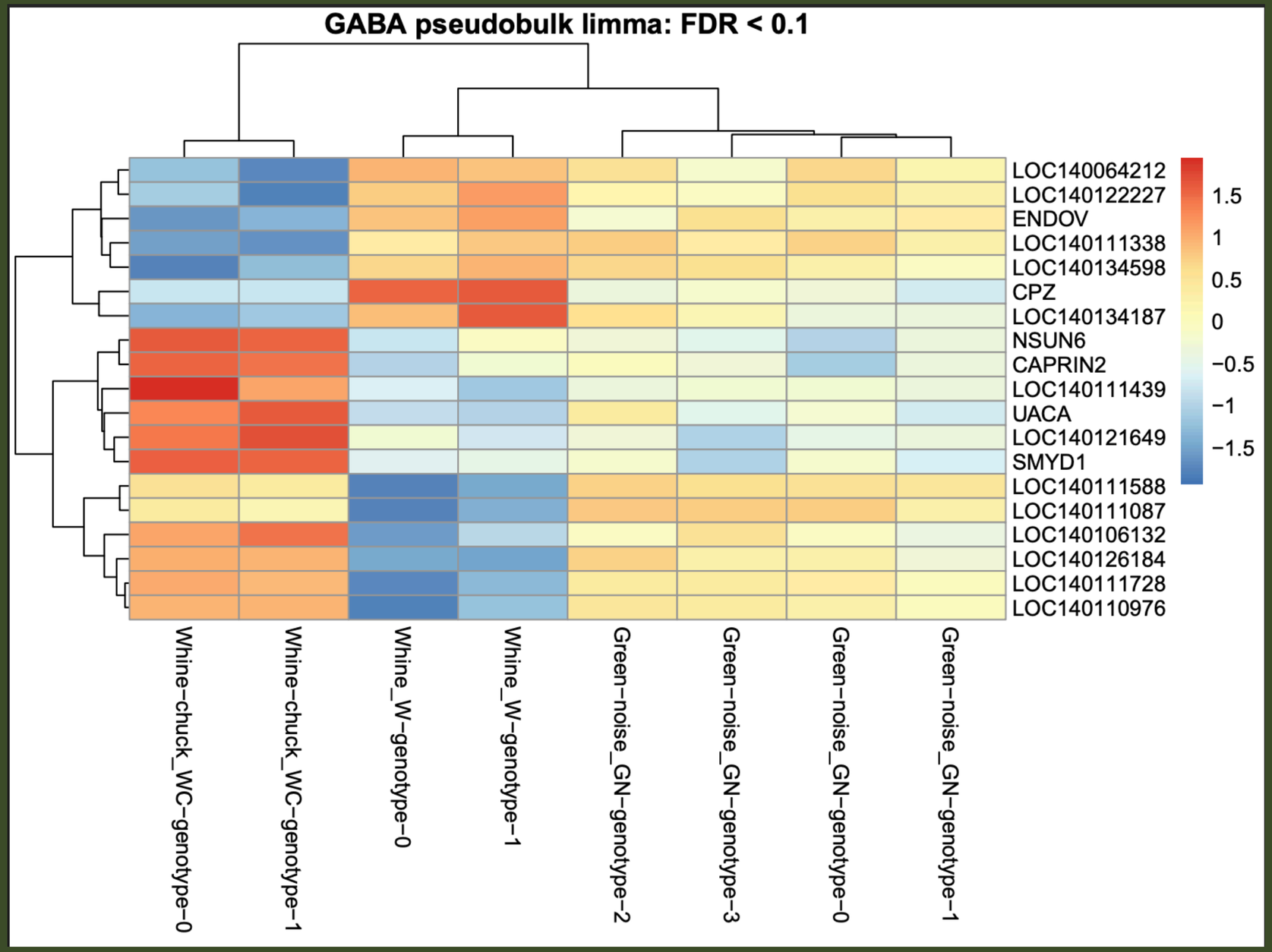
GABAergic Neurons: WC vs. W DEG Heatmap



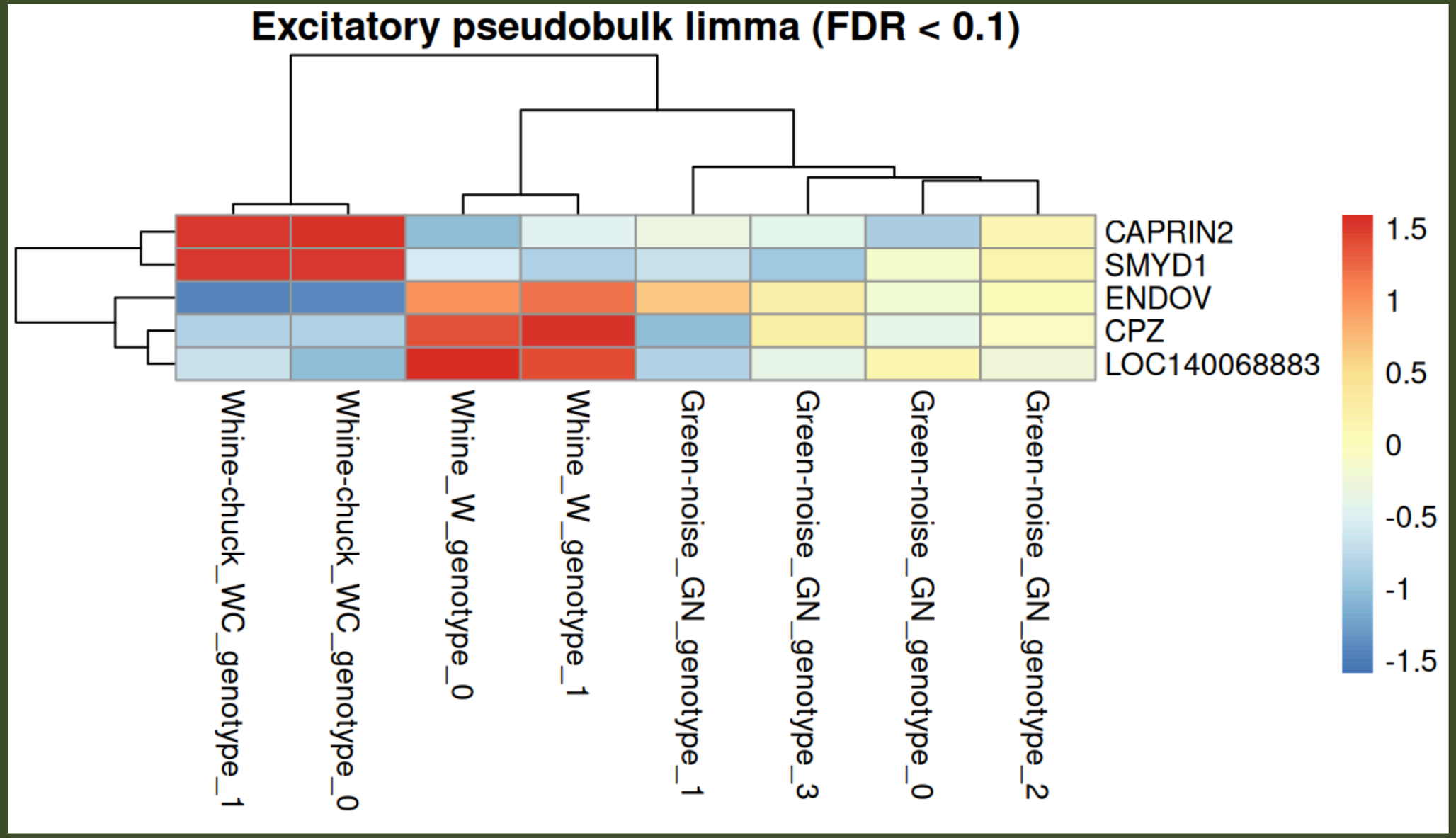
Excitatory Neurons: WC vs. W DEG Heatmap

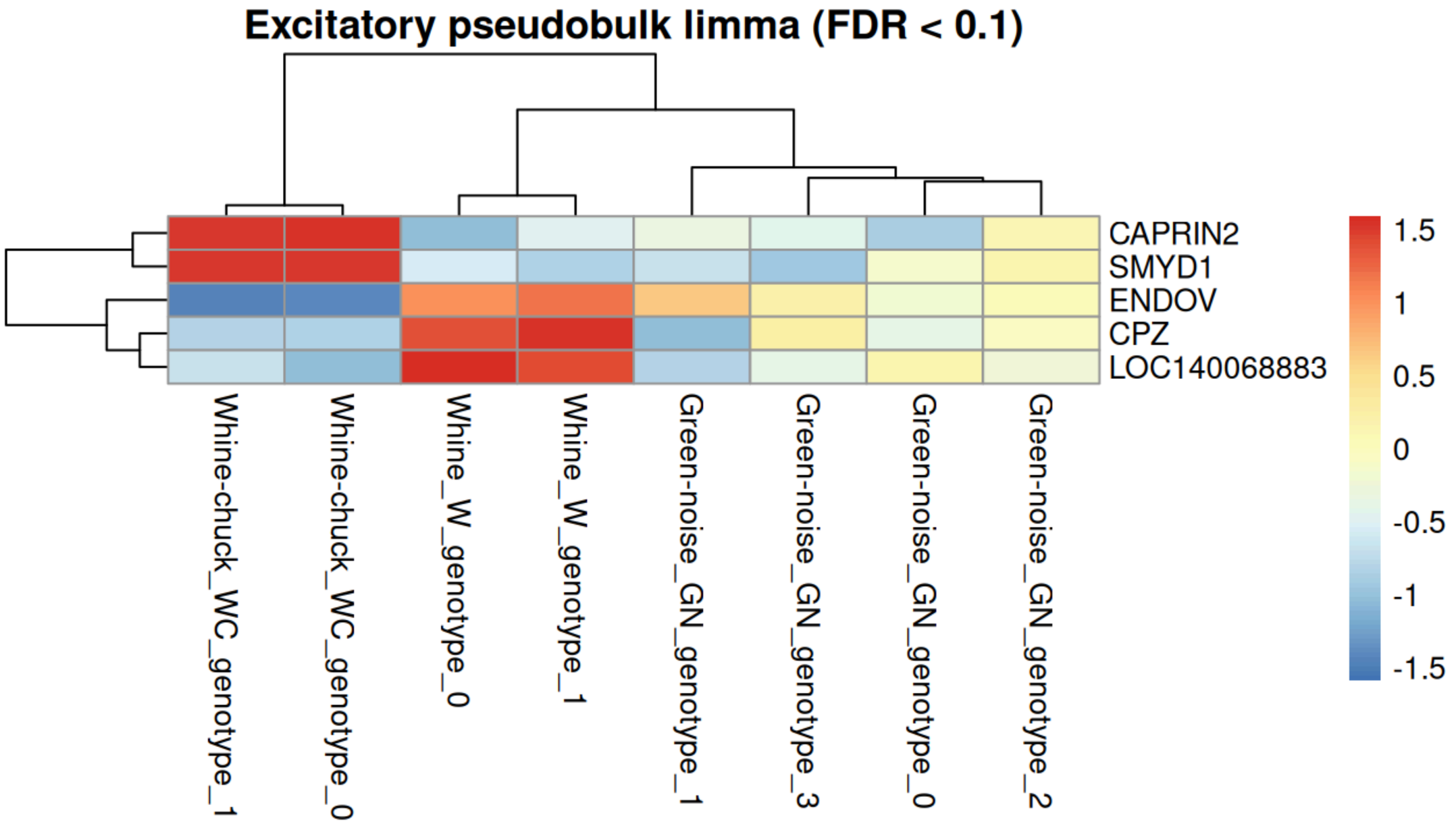


GABAergic pseudobulk limma (Souporcell): WC vs W

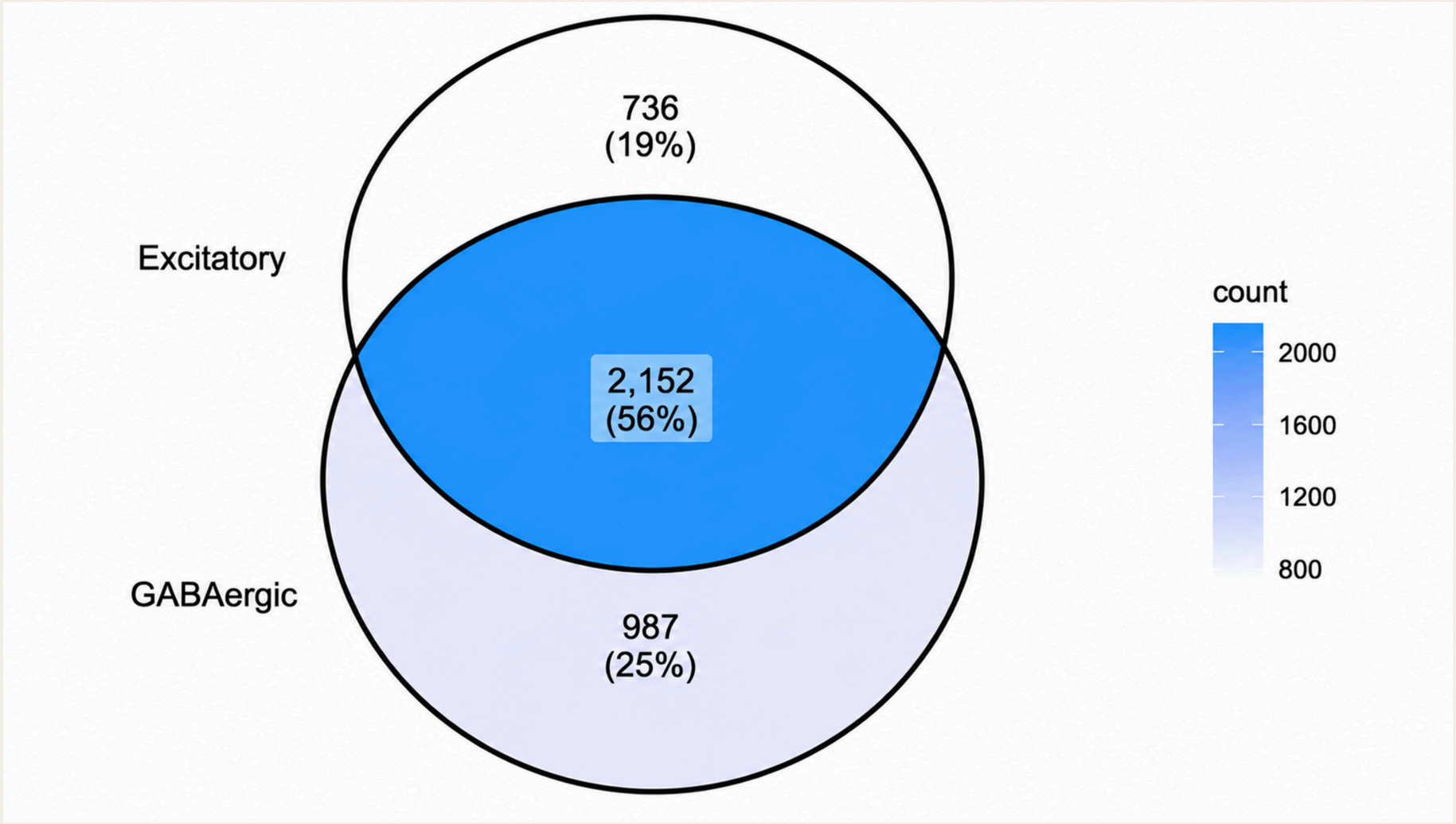


Excitatory pseudobulk limma (Souporcell): WC vs W




Excitatory pseudobulk limma (FDR < 0.1)
CAPRIN2
SMYD1
ENDOV
CPZ
LOC140068883
Whine-chuck_WC_genotype_1
Whine-chuck_WC_genotype_0
Whine_W_genotype_0
Whine_W_genotype_1
Green-noise_GN_genotype_1
Green-noise_GN_genotype_3
Green-noise_GN_genotype_0
Green-noise_GN_genotype_2
1.5
1
0.5
0
-0.5
-1
-1.5

DEG Overlap: W vs. WC



Top DEGs - Excitatory and GABAergic

Gene	GABAergic avg_log2FC	Excitatory avg_log2FC	p_val_adj	Significance
CAPRIN2	7.9168650	7.9012290	0	Upregulated in WC
NRTN	3.8112611	3.6743100	0	Upregulated in WC
SLC22A2	3.5790562	3.6112460	0	Upregulated in WC
SHISA8	0.8722207	-	0	Upregulated in WC
FAM83E	-	-4.7856360	0	Downregulated in WC
LOC140111338	-	-3.8069410	0	Downregulated in WC
MFSD14A	-1.3353053	-	0	Downregulated in WC
LOC140111485	-8.0226801	-8.4950940	0	Downregulated in WC
MEI1	-6.4348536	-6.3956200	0	Downregulated in WC
LOC140108663	-2.7548216	-2.8908380	0	Downregulated in WC
GPR155	-2.7726693	-2.7109180	0	Downregulated in WC
LOC140119761	-1.6292934	-1.6830780	0	Downregulated in WC

GABAergic Neurons (Upregulated DEGs)

CAPRIN2

- Drives synaptic plasticity, allowing the brain to rapidly form new connections.
- Remodels inhibitory circuits to help control and balance the incoming sound signals so the brain doesn't short-circuit.

NRTN

- Protects heavily taxed brain cells from dying during intense electrical firing.
- Keeps the inhibitory "brake" cells healthy and fully functional while the rest of the brain is aggressively rewiring.

SLC22A2

- Clears out metabolic byproducts and shuttles essential neurotransmitters.
- Manages the intense metabolic load required to maintain the brain's delicate balance of excitation and inhibition.

SHISA8

- Adjusts exactly how strongly the neuron responds to incoming electrical signals.
- Directly fine-tunes the brain's "brakes" to keep the massive surge of excitatory signals perfectly balanced and controlled.

Excitatory Neurons (Upregulated DEGs)

CAPRIN2

- Remodels circuits in real-time to quickly process the complex "Chuck" sound and drive the frog's physical response.

NRTN

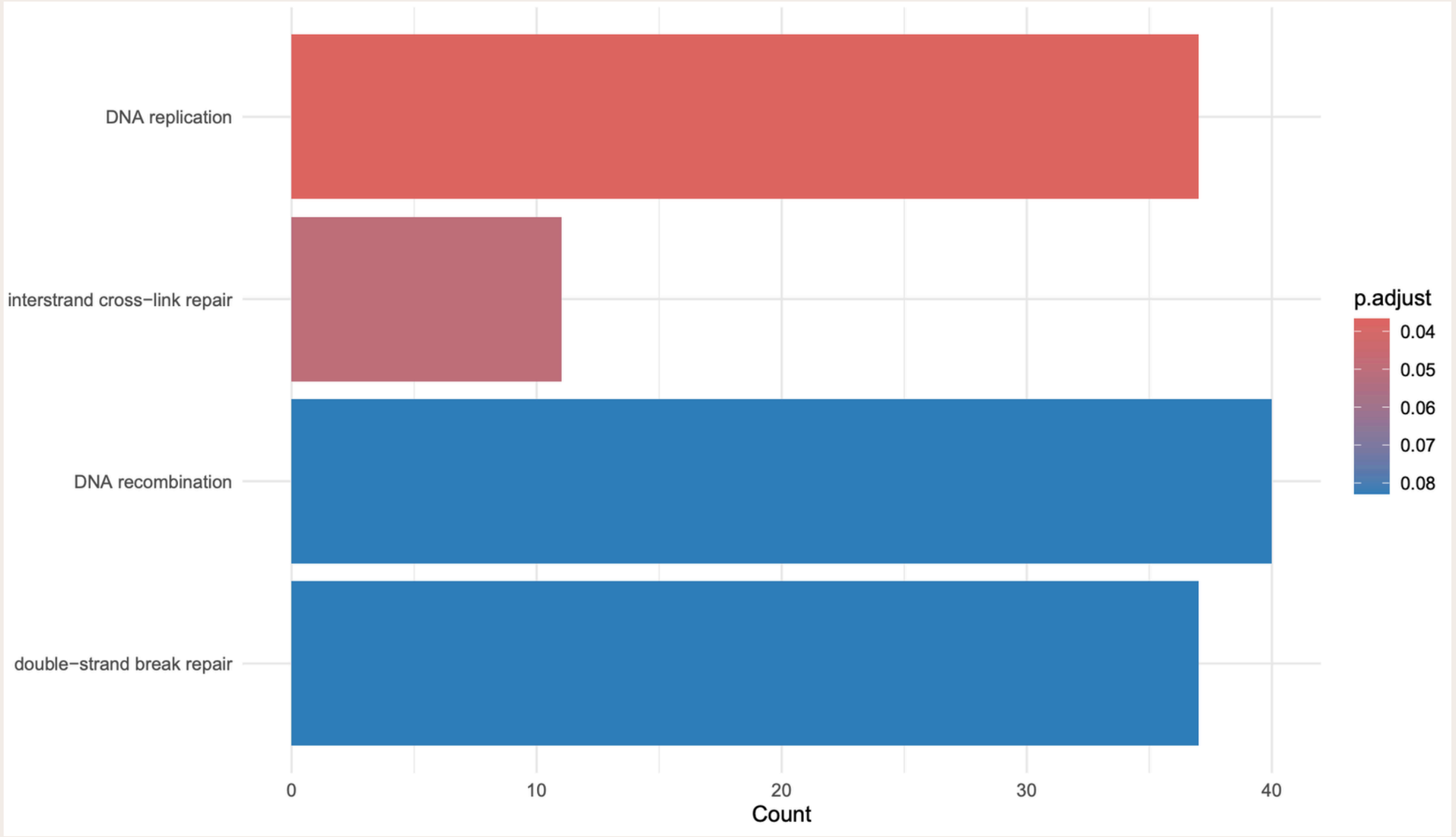
- Ensures the "gas pedal" neurons survive the massive energy surge required to process the complex mating call.

SLC22A2

- Keeps the internal environment of the active neuron completely stable while it rapidly fires and builds new connections.

GO ENRICHMENT

GO ENRICHMENT - GABAergic and Excitatory



ENRICHMENT RESULTS

1 DNA Replication

- Directly provides the resources needed to build new synaptic connections and process the complex "Chuck" sound.

2 Interstrand cross-link repair

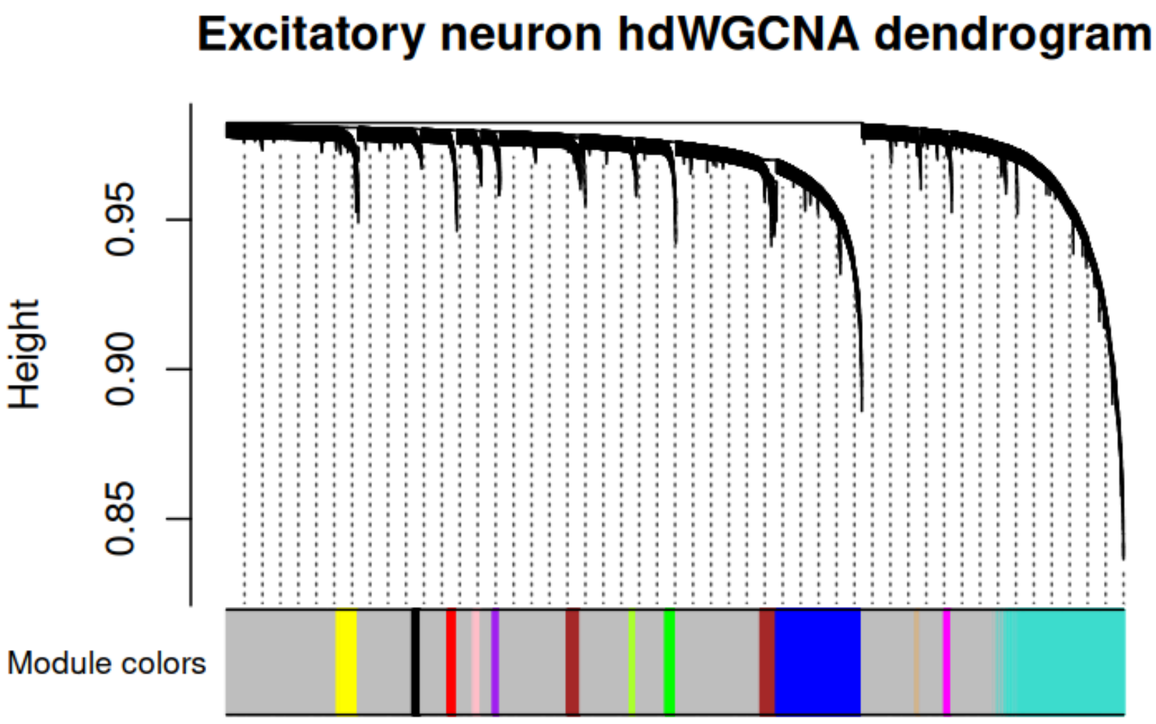
- Quickly untangles and fixes structural knots in the DNA, ensuring the heavily taxed brain cells survive the process.

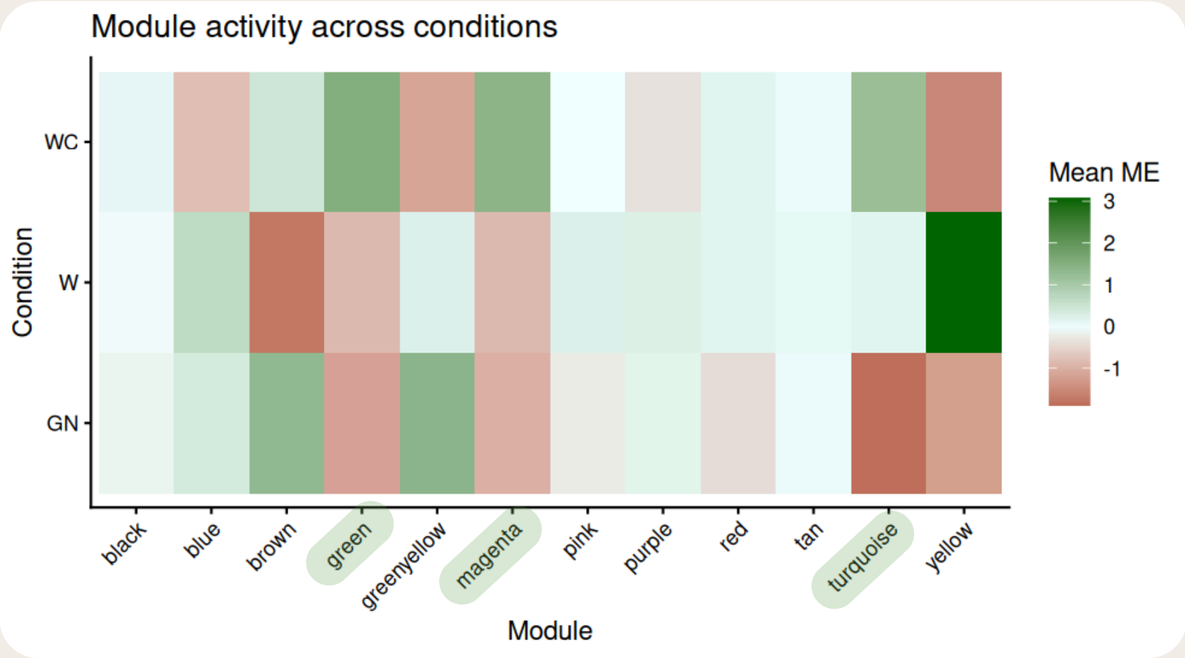
6

WGCNA Analysis

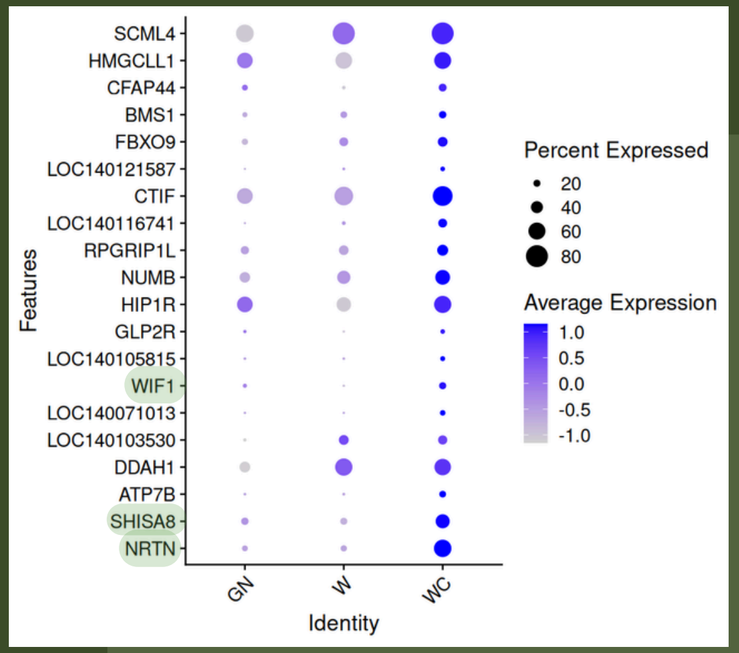
hdWGNCA

EXCITATORY NEURONS

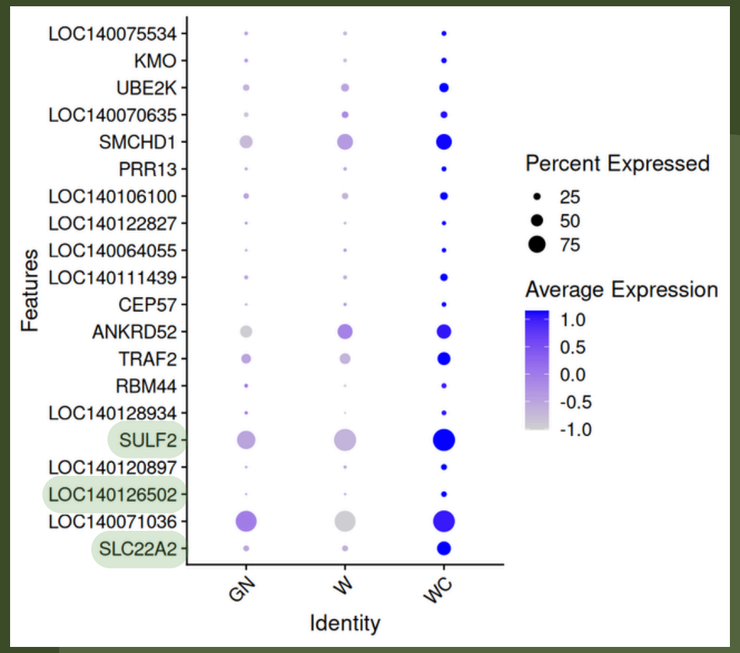




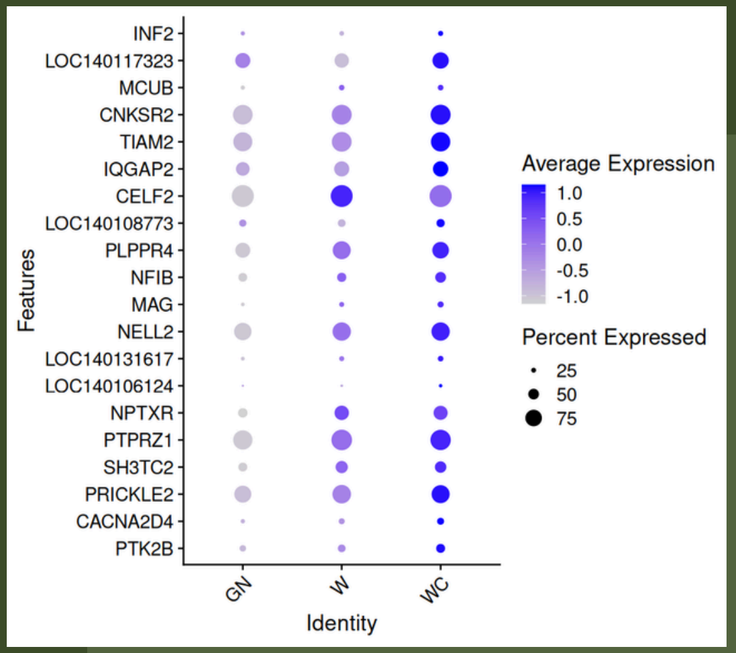
HUB GENES



GREEN MODULE



MAGENTA MODULE

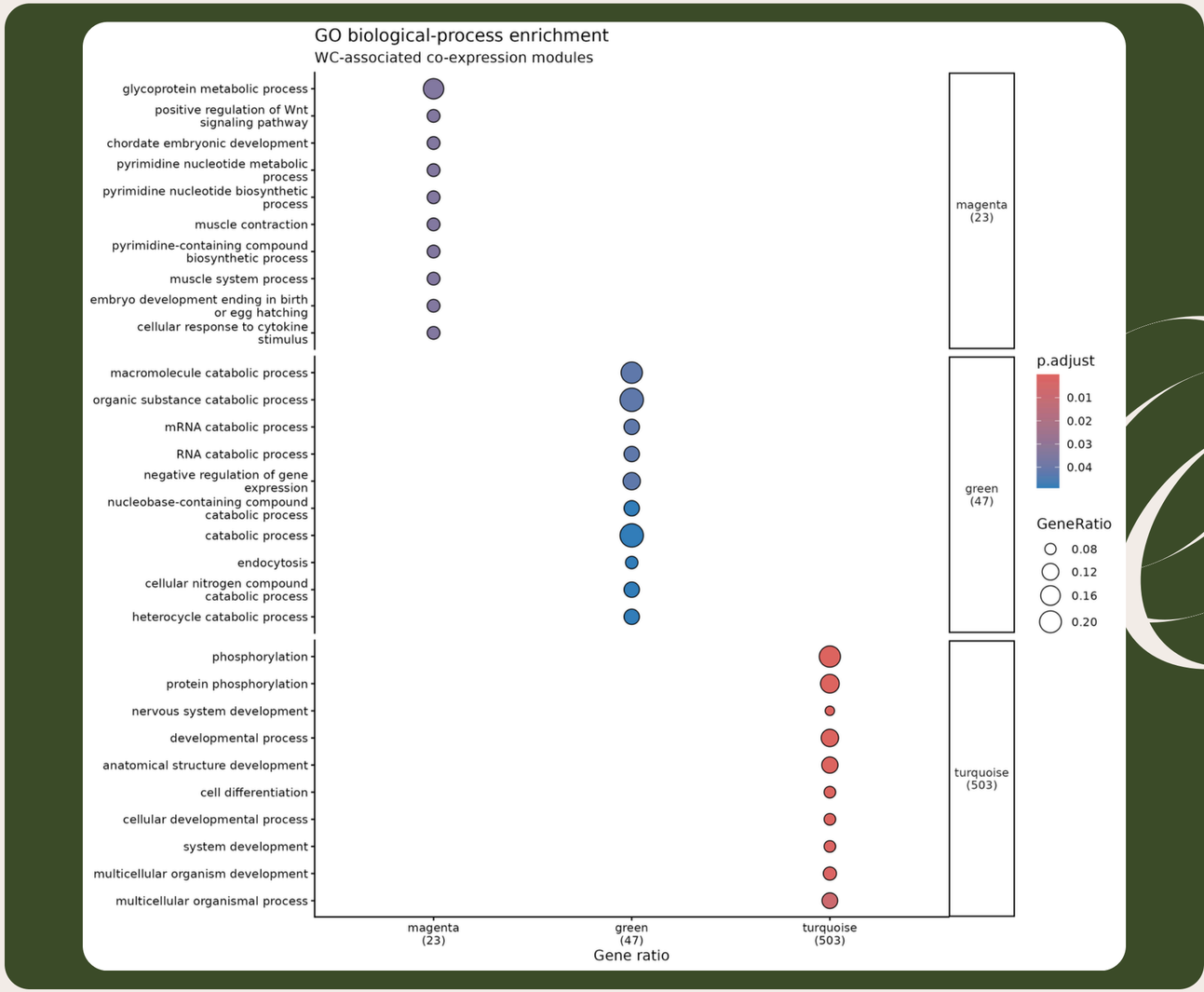


TURQOUISE MODULE

Highlighted within each module is genes that also showed up in our DEG list

GO ENRICHMENT

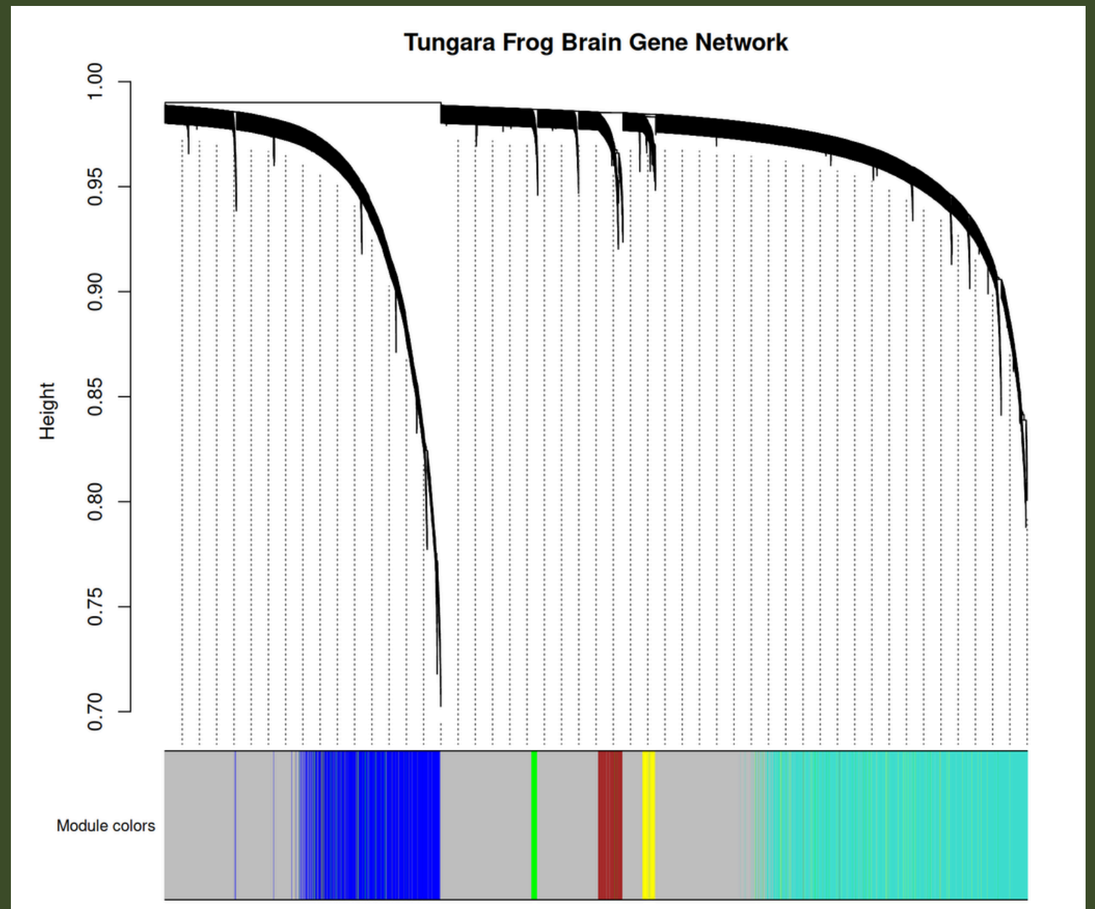
- **Magenta module**: May relate to cell signaling and changes in how cells grow or adapt, which could be connected to neural plasticity.
- **Green module**: Mainly represents catabolic processes, meaning the breakdown and recycling of RNA, proteins, and other cell materials.
- **Turquoise module**: Represents broader cell development and communication processes, including phosphorylation, nervous system development, and cell differentiation.

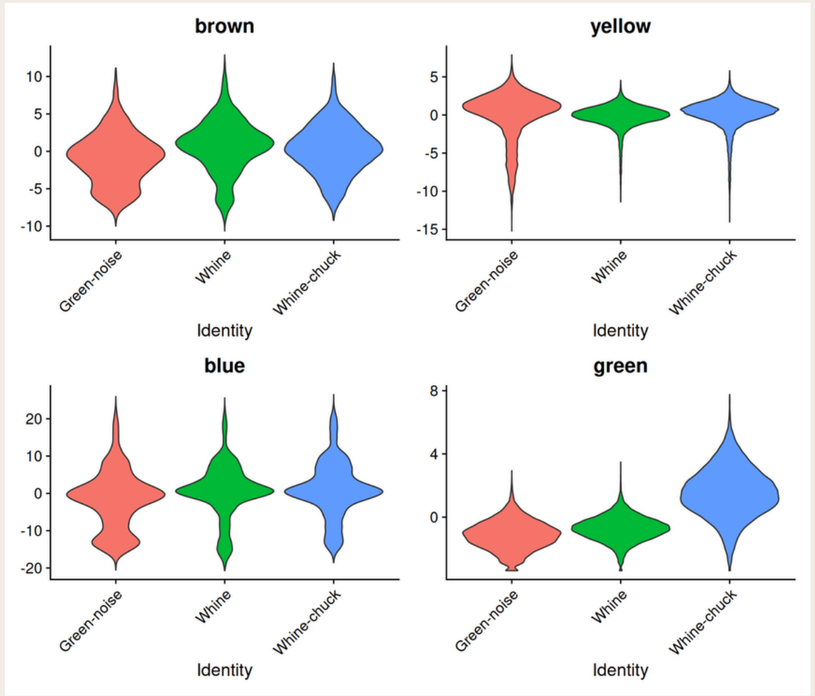


hdWGNCA

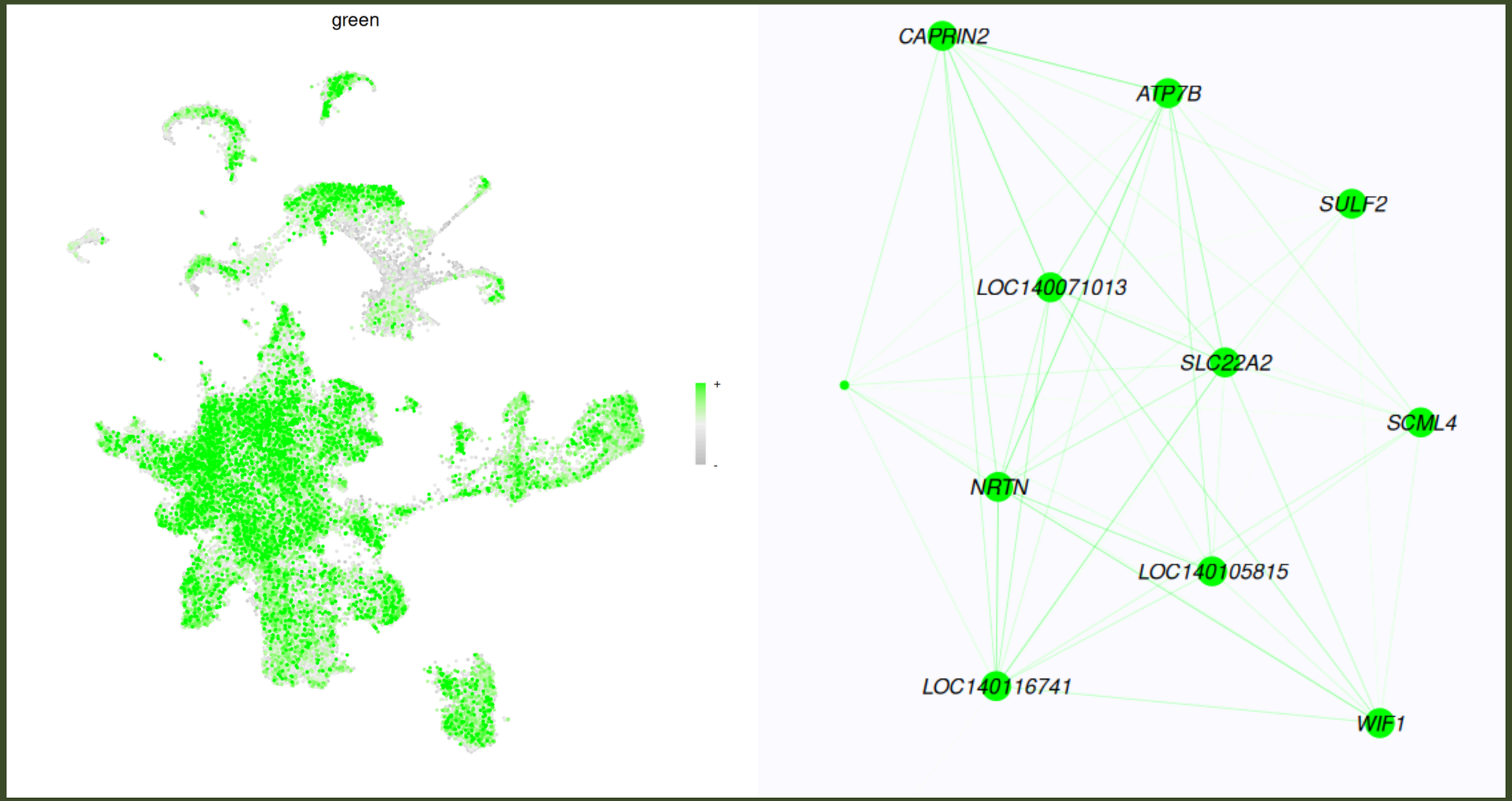
Gabaergic NEURONS

grey	blue	turquoise	green	yellow	brown
50.24	14.36	28.55	1.63	2.42	2.79

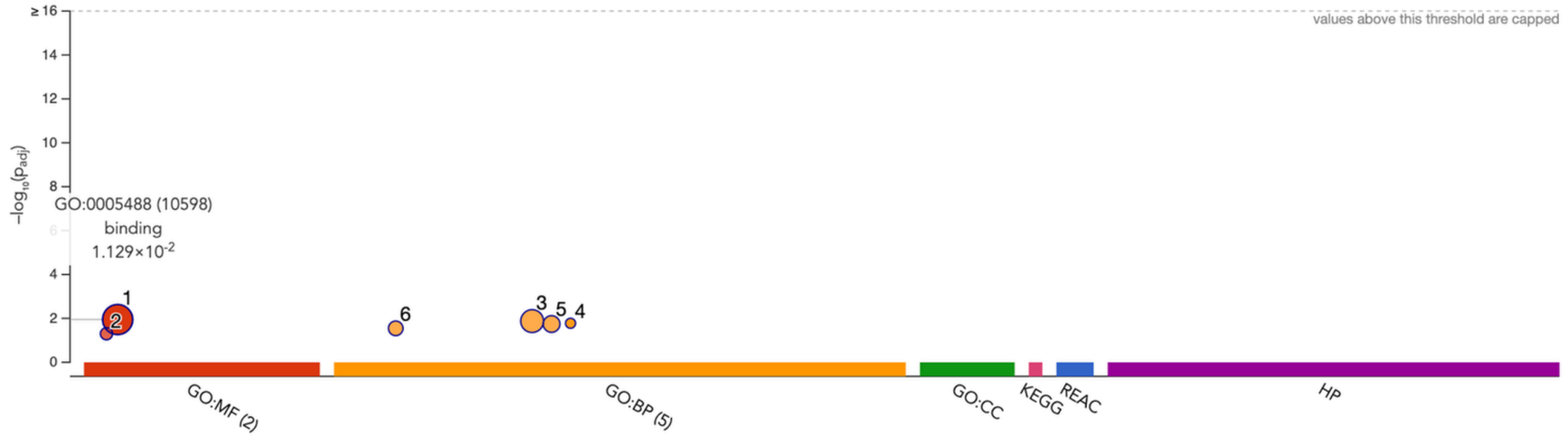




Green hub


green
+
-
CAPRIN2
ATP7B
SULF2
LOC140071013
SLC22A2
SCML4
NRTN
LOC140105815
LOC140116741
WIF1

Gprofiler

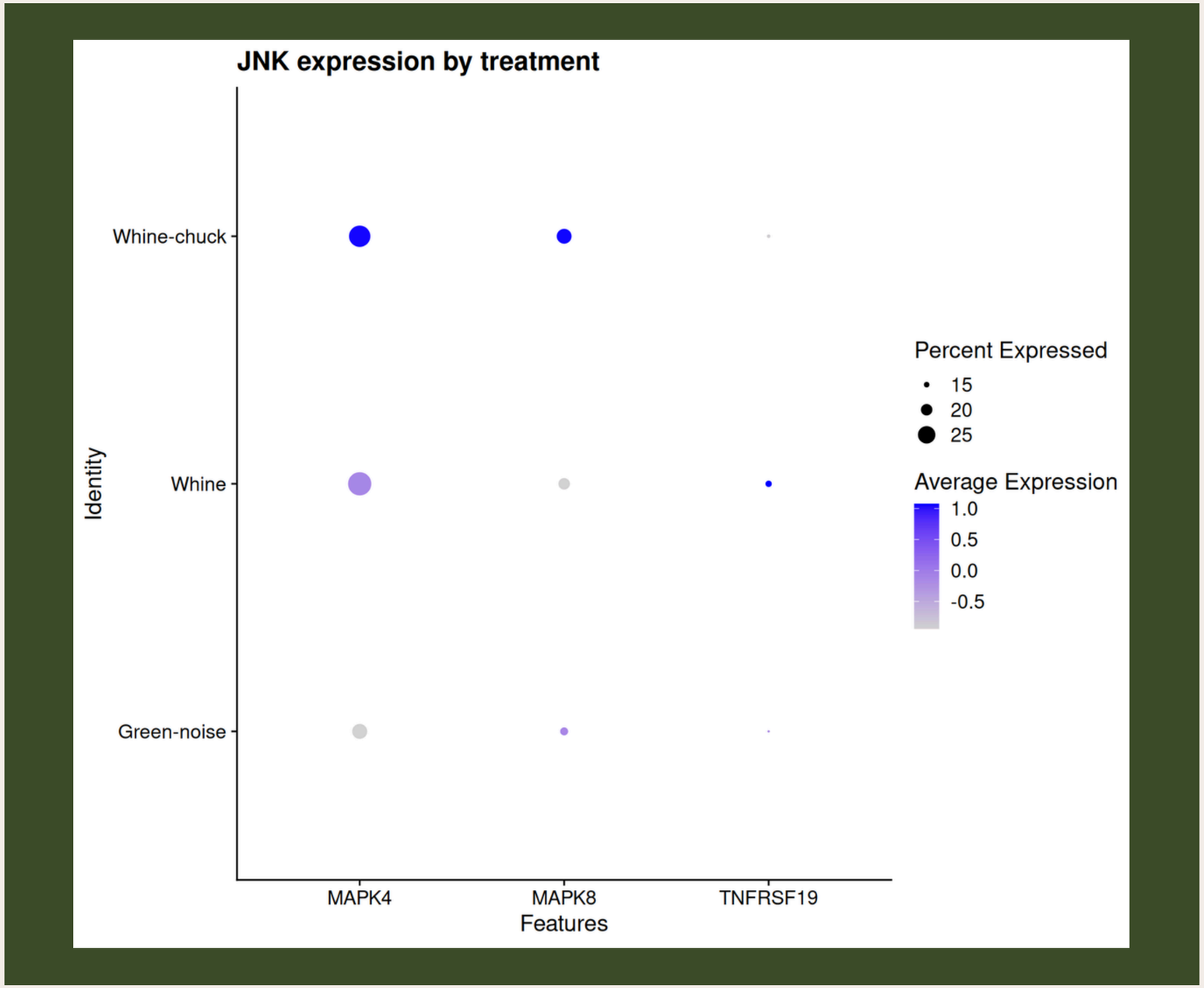


ID	Source	Term ID	Term Name	p_{adj} (query_1)
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2	GO:MF	GO:0004705	JUN kinase activity	4.995×10^{-2}
3	GO:BP	GO:0035556	intracellular signal transduction	1.329×10^{-2}
4	GO:BP	GO:0043697	cell dedifferentiation	1.672×10^{-2}
5	GO:BP	GO:0042127	regulation of cell population proliferation	1.803×10^{-2}
6	GO:BP	GO:0007254	JNK cascade	2.838×10^{-2}

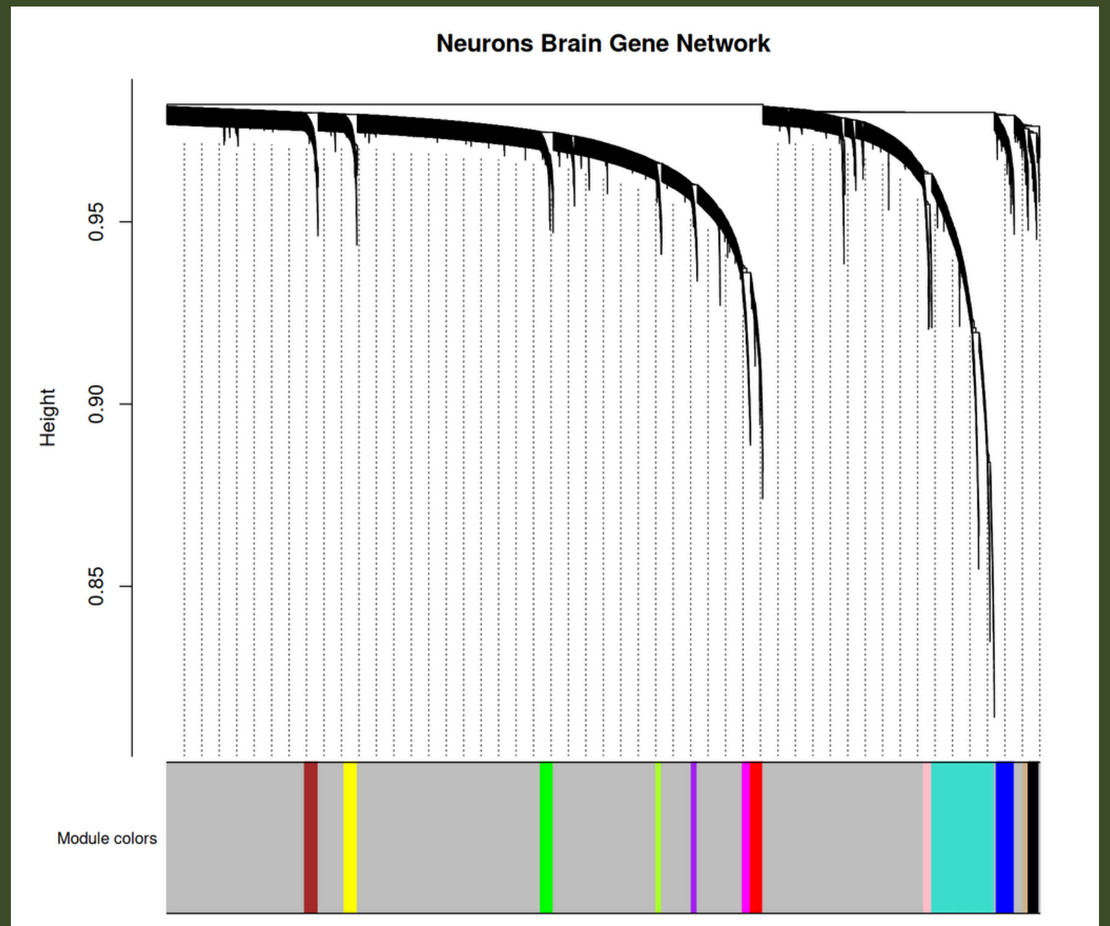
gProfiler_xtropicalis_2026-06-22_19-32-

source	term_name	term_id	highlighted	adjusted_p_value	negative_log10_of_adjusted_p_value	term_size	query_size	intersection_size	effective_domain_size	intersections
GO:MF	binding	GO:0005488	TRUE	0.011292346907130322	1.9472157885539678	10598	73	64	15795	XP_031750900,X
GO:MF	JUN kinase activity	GO:0004705	TRUE	0.04995257159178639	1.3014421490921628	5	73	2	15795	XP_004910415,X
GO:BP	intracellular signal transduction	GO:0035556	TRUE	0.013294676864923555	1.8763222139913889	473	61	11	12163	XP_004910415,X
GO:BP	cell dedifferentiation	GO:0043697	TRUE	0.016717589778886207	1.7768263358193932	2	61	2	12163	NP_988885,XP_
GO:BP	regulation of cell population proliferation	GO:0042127	TRUE	0.018031551555557503	1.7439669020615334	35	61	4	12163	NP_988885,XP_
GO:BP	JNK cascade	GO:0007254	TRUE	0.028382618267177383	1.546947543813796	14	61	3	12163	XP_004910415,X

Gprofiler

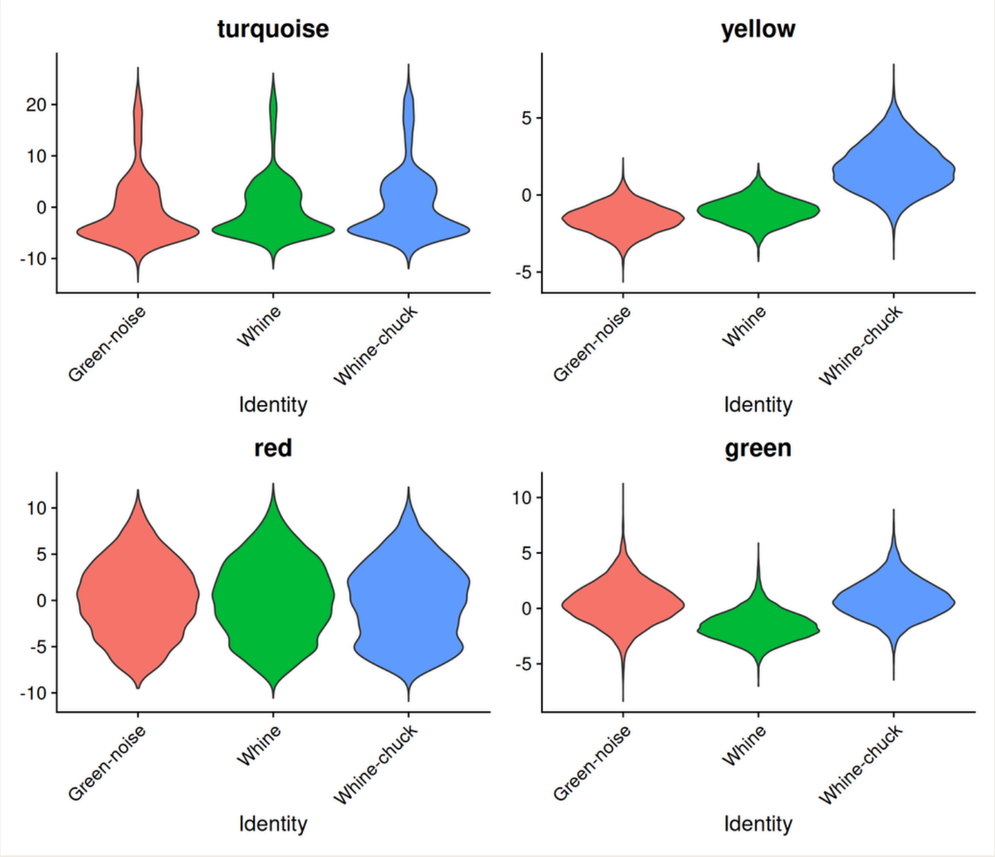

JNK expression by treatment
Whine-chuck
Whine
Green-noise
Identity
MAPK4
MAPK8
TNFRSF19
Features
Percent Expressed
15
20
25
Average Expression
1.0
0.5
0.0
-0.5

grey	turquoise	yellow	red	green	greenyellow	blue	magenta	tan	pink	purple
82.98	7.45	1.59	1.48	1.54	0.68	2.11	0.94	0.59	1.01	0.69
black	brown									
1.29	1.61									

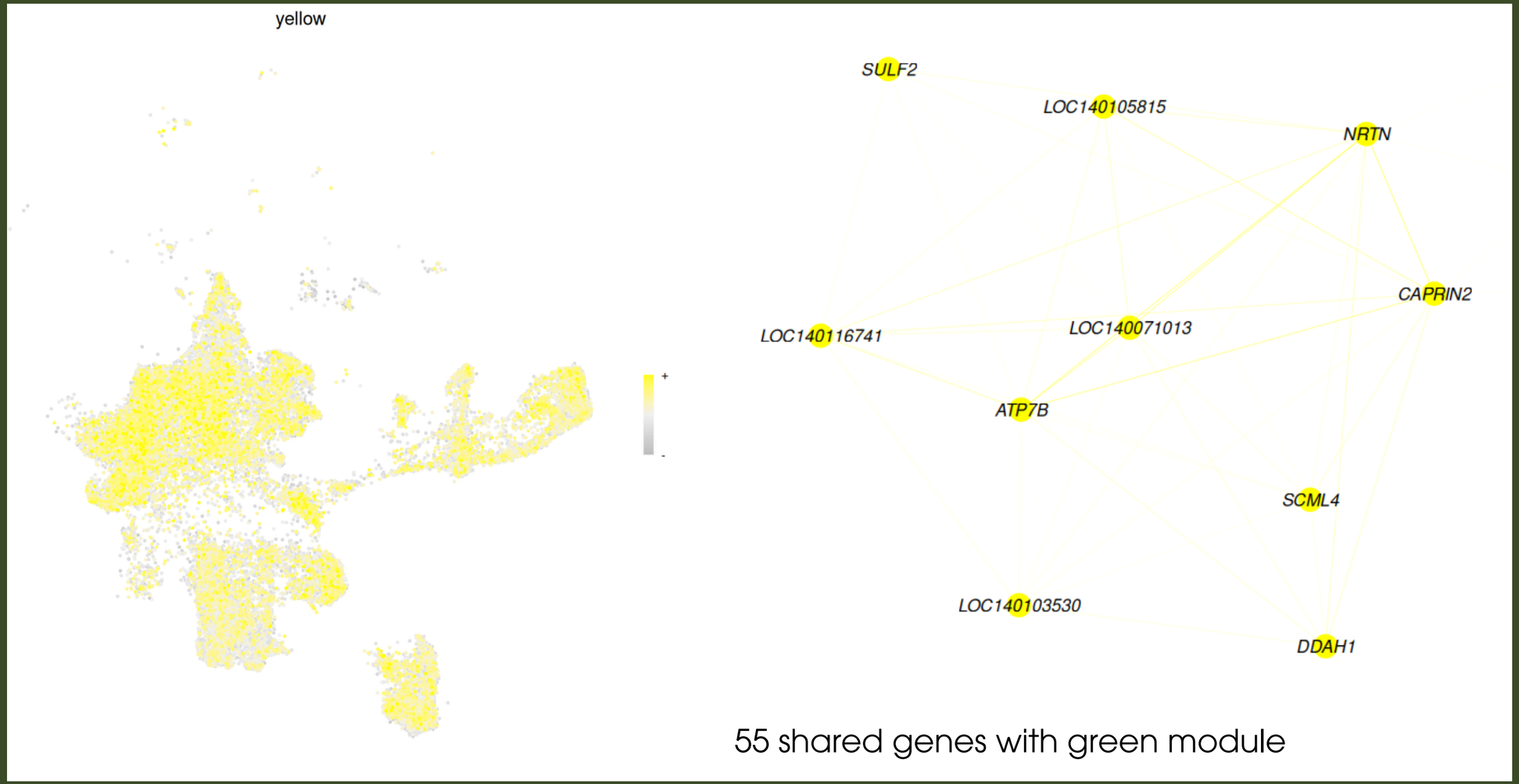


hdWGNCA

ALL NEURONS



Yellow hub



55 shared genes with green module

6

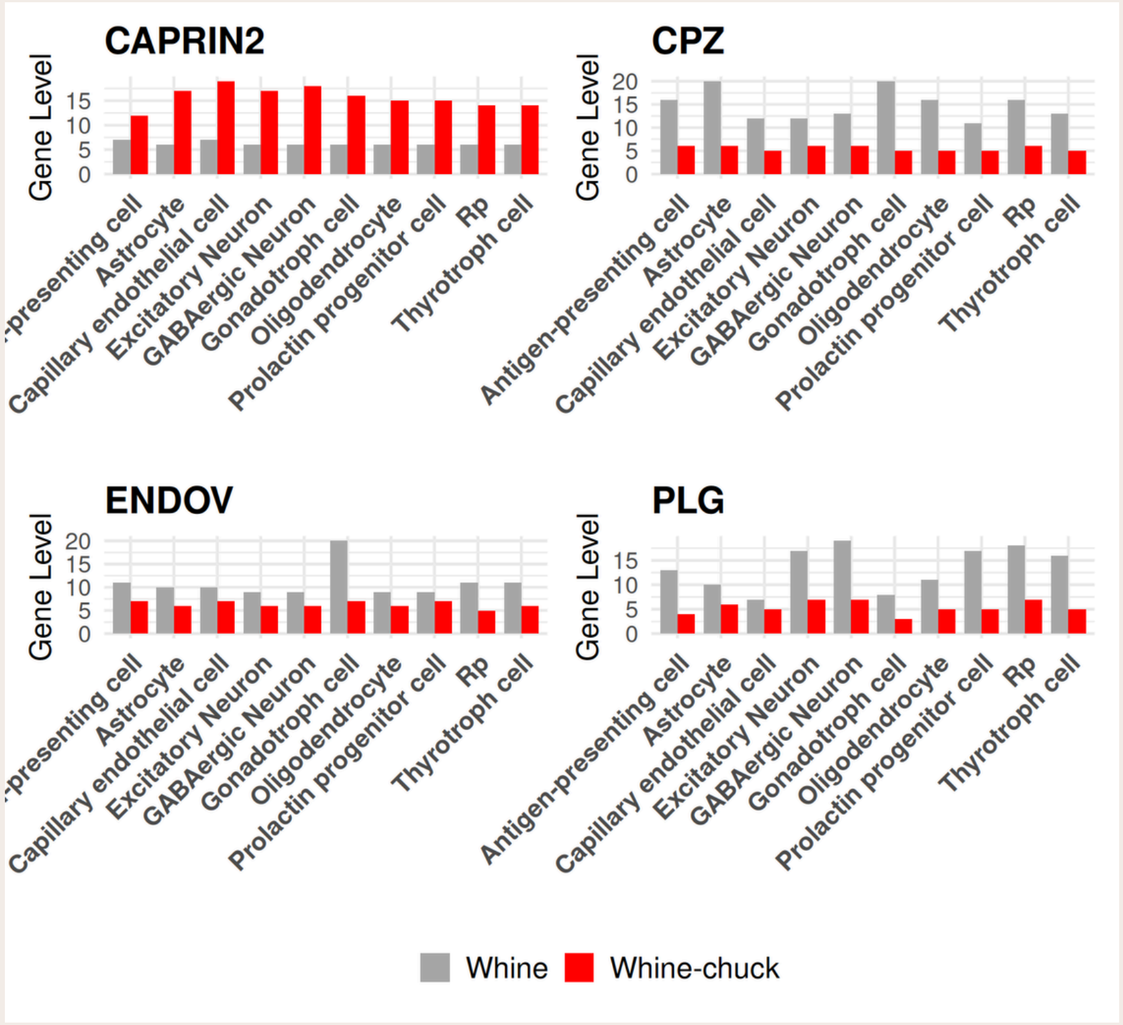
Conclusion

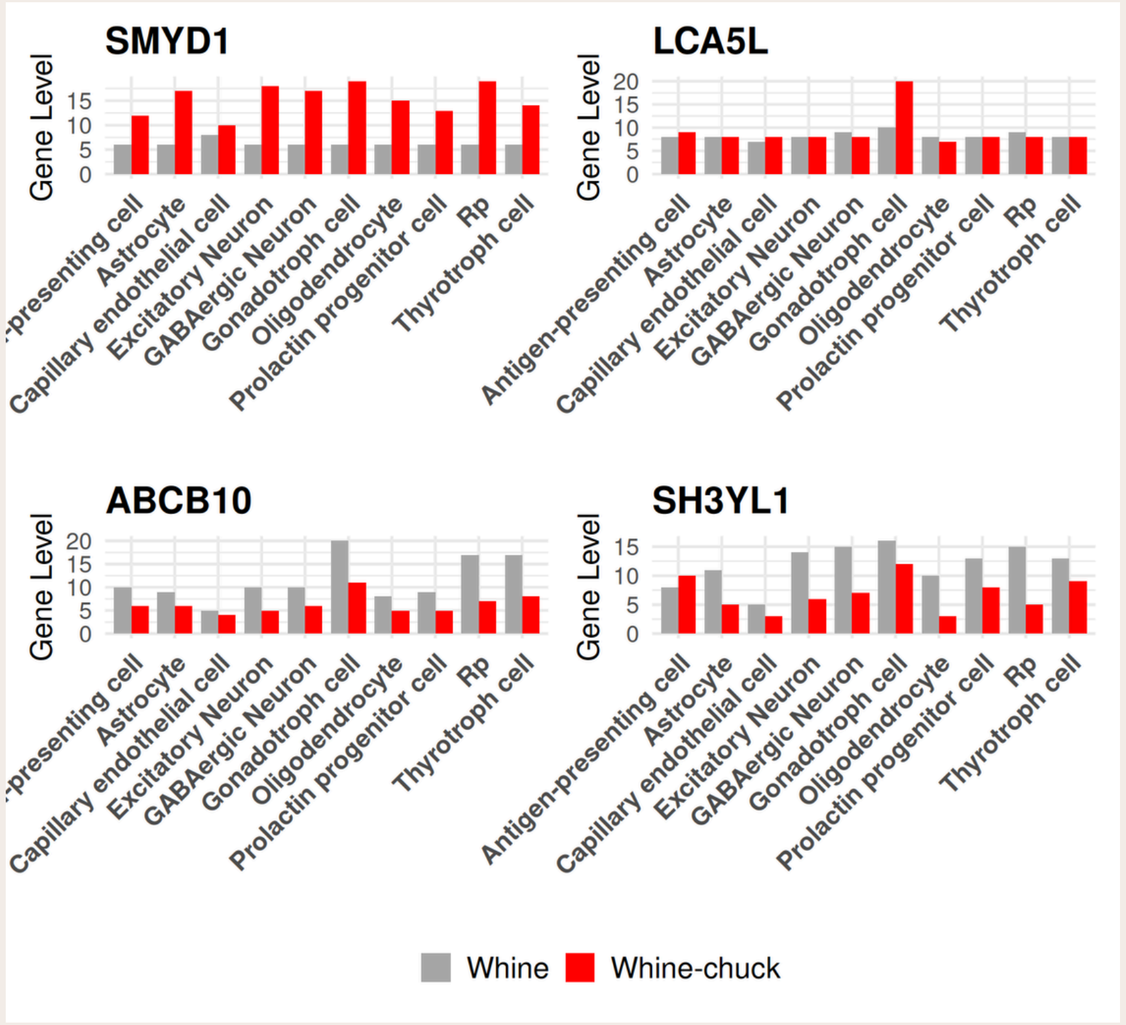
Conclusion



- Annotated brain cell populations in the túngara frog brain
- Whine-chuck calls induce strong transcriptional changes in both excitatory and GABAergic neurons
- Pseudobulk (Souporcell + limma) identified reproducible candidate genes, including CAPRIN2, NRTN, and SLC22A2
- WGCNA revealed gene networks involved in neural plasticity and cell signaling
- These findings provide insight into the molecular basis of mate-choice behavior

Brain wide gene high expresion



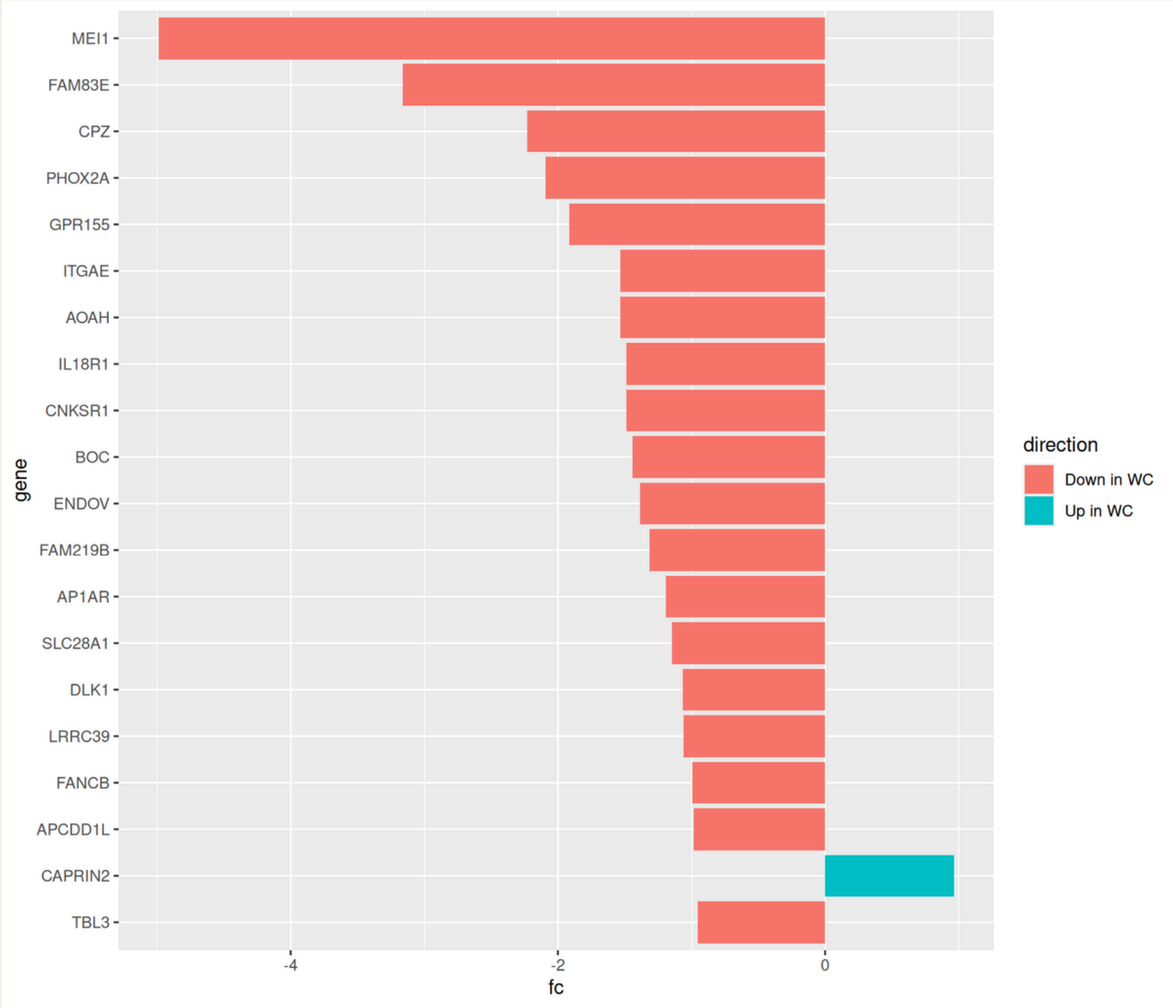


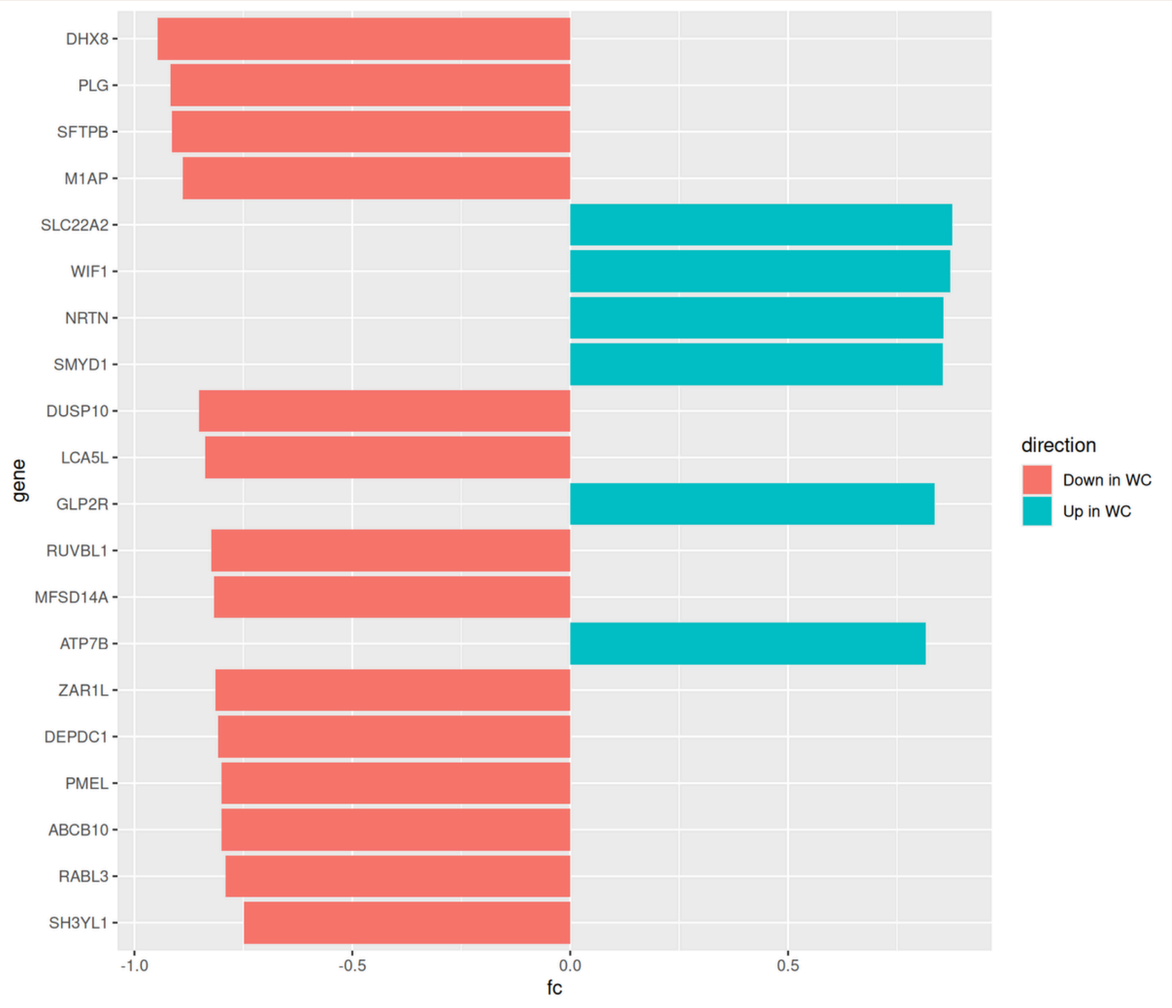


Limitations

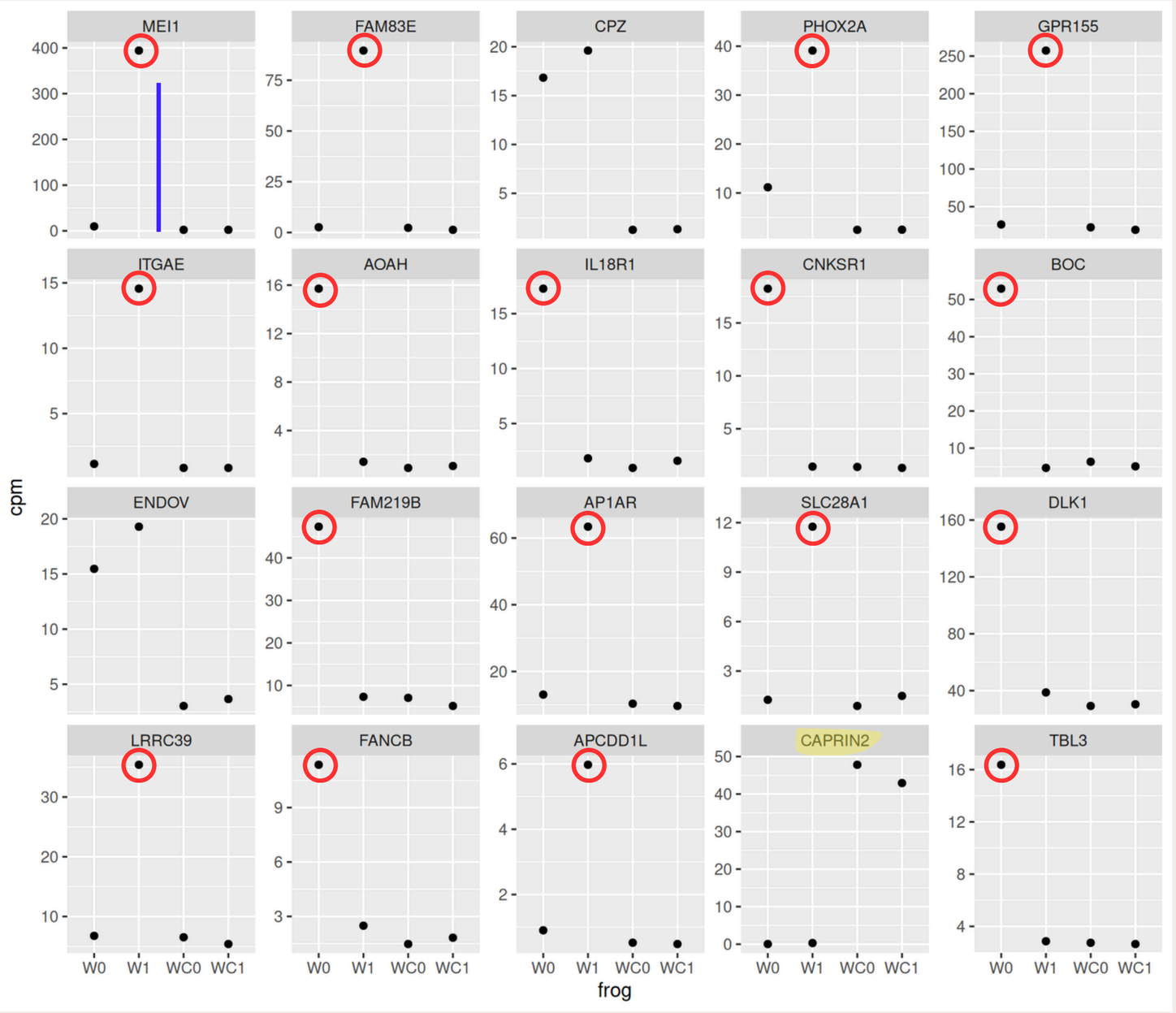
- Small sample sizes for the whine and whine-chuck groups, with only two biological samples in each condition.
- Some XCL markers used older Xenopus gene names, preventing several genes from being mapped and included despite multiple retrieval attempts.
- In this dataset a significant portion of genes are LOC genes
- The marker set list could be further improved but would require extensive review of literature to build

Genes with highest absolute logfold2change





Genes split by genotypes in MPC



SOURCES

- Liao, Y., Ma, L., Guo, Q. et al. Cell landscape of larval and adult Xenopus laevis at single-cell resolution. Nat Commun 13, 4306 (2022). https://doi.org/10.1038/s41467-022-31949-2
- Page, Rachel A., and Ximena E. Bernal. “Túngara Frogs.” Current Biology, vol. 16, no. 23, Dec. 2006, pp. R979–R980, https://doi.org/10.1016/j.cub.2006.10.046.
- Dwortz MF, Hofmann HA, Curley JP. Social processing in the amygdala: single-nucleus RNA-sequencing reveals distinct neuronal responses to dominant and subordinate cues. Scientific Reports (2026). https://doi.org/10.1038/s41598-026-52921-w

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THANK YOU